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[Exploring Weakly Supervised Multitask Learning for Cross-Domain Spam Detection in News Media]

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ABSTRACT

The growth of online news media has increased the spread of spam and misleading content, raising concerns about the credibility and trustworthiness of information. Detecting spam news articles is especially difficult across different domains, where content covers various categories like politics, health, business, and entertainment where annotated datasets are limited. Manually labelling large sets of news articles costs a lot and can be subjective, which creates a need for automated methods that can learn effectively from noisy or weak supervision.

This dissertation looks into the issue of cross-domain spam news detection with weak supervision, aiming to improve performance through multitask learning. The main idea is that adding extra tasks, like predicting the category and source of a news article, can help stabilize learning and improve spam detection even when label quality is low. This is important academically, given the rising interest in weak supervision and multitask models in natural language processing, and it is also crucial for enhancing the reliability of automated moderation tools across various news platforms.

The proposed method combined weak labelling functions with deep learning models. Weak labels were produced using heuristic and rule-based functions, such as identifying clickbait patterns, excessive punctuation, unusual sources, mismatches between titles and content. These functions provided noisy but scalable spam labels along with a confidence score for each instance. Based on this dataset, a series of experiments were conducted: (1) Traditional TF-IDF and Logistic Regression as a baseline, (2) Single-Task Transformer model (RoBERTa) for spam classification, (3) Multitask learning with a shared encoder and two outputs (spam and category); and (4-5) Advanced multitask models that included a source-prediction output, evaluated with and without metadata tokens.

The results showed a clear improvement in performance. Classical models reached modest F1 scores of about 0.41. Transformer baselines improved precision but had limited recall, with an F1 score around 0.49 to 0.54. Adding multitask learning with category prediction led to steady increases in spam F1 scores of about 0.55 to 0.57, supporting the idea that additional signals help the main task. While an upper-bound experiment with metadata tokens raised performance (F1 about 0.68), the cleaner multitask model without metadata tokens gained more modest but methodologically sound results with an F1 of around 0.54, highlighting the challenge of creating meaningful auxiliary tasks. Category prediction was very accurate, scoring above 0.85, while source prediction was tougher, showing the noisy nature of source metadata.

The main contribution of this work is a thorough evaluation of weakly supervised multitask learning for cross-domain spam news detection, illustrating both the potential and the limitations of auxiliary tasks in noisy conditions. The findings suggest that while multitask learning can boost recall and provide regularization, its effectiveness largely relies on the usefulness of auxiliary signals. This contributes to the broader literature by clarifying the conditions under which weakly supervised multitask approaches can lead to improvements. Future work should explore richer metadata features and advanced weak supervision frameworks to further enhance cross-domain spam detection.

HIGHLIGHTS

- Weak supervision generates scalable spam labels for news classification.
- Multitask learning improves spam detection across multiple news domains
- Transformer models perform better than traditional TF-IDF benchmarks in spam detection.
- Auxiliary category prediction improves spam F1 compared to single-task models
- Metadata tokens caused label leakage. It was removed to ensure a fair evaluation
- Findings clarify limits of multitask learning with noisy weak supervision

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I certify that the work presented in the dissertation is my own unless referenced

Signature.....Preethi Sivaradjou.....

Date.....08/09/2025.....

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CHAPTER 1: INTRODUCTION

Online news and social media have undergone a significant shift in how we produce and consume information. While there are indeed channels that allow for the quick sharing of credible information, they often come hand-in-hand with spam and misleading articles designed to trick readers or grab clicks. These spammy news pieces, usually wrapped in clickbait or poorly written content, significantly harm the integrity of information and contribute to a world filled with misinformation. Therefore, being able to identify this type of content is crucial for protecting users and maintaining the quality of information shared online.

1.1 Background

Traditional approaches to recognizing spam or fake news are nearly entirely supervised learning approaches, in which the classifier is trained using human-annotated collections of data. Such excellent quality annotated collections are expensive to prepare, narrowly focused on limited domains, and do not transfer across new categories or sources *(Shu, Mahudeswaran, and Liu, 2019)*. To overcome this challenge, more recent literature has examined weak supervision, where noisy or heuristic labelling functions are utilized to create labels for the training data on a large scale. *Shu et al. (2020)*, for example, demonstrated that a collection of weak social indicators can be combined to recognize fake news early on. In contrast, *Özgöbek, Bayrak, and Aydin (2022)* demonstrated that content-based weak supervision is a cost-effective alternative to human labelling, which is often a prerequisite for supervised approaches. Similarly, *Wang et al. (2020)* described reinforcement learning techniques for enriching weakly labelled information for fake news detection tasks. These papers highlight the promise of weak supervision for scalable detection of spam news.

Meanwhile, multitask learning (MTL) has proven to be a successful paradigm in natural language processing, enabling models to learn a shared representation of multiple related tasks. In the case of misinformation, auxiliary tasks like recognizing the emotion of the post or detecting novelty have been found to enhance classification *(Kumari et al., 2021)*. In a similar line of work to detecting media bias, multitask transformer models have been successful in better generalisation across datasets *(Spinde, Vassiliou and Schenck, 2022)*. In the case of spam detection also, multitask methods have been tried and tested in the setting of the problem of email filtering and detecting fake reviews, where auxiliary information has been found to enhance performance *(Chelliah, Manikandan, and Subramanian, 2021; Melleng, Aouedi and Ezzikouri, 2023)*.

Despite these advances, a gap remains at the intersection of weak supervision and multitask learning for cross-domain spam detection. Most prior studies investigate either weakly supervised fake news detection *(Shu, Mahudeswaran and Liu, 2019; Wang et al., 2020)* or multitask models for related NLP problems (Kumari et al., 2021), but few combine the two. Moreover, existing work often focuses on single-domain datasets or benchmark corpora, which limits its applicability to real-world news streams where spam spans multiple categories, such as politics, health, business, and entertainment.

This dissertation addresses that gap by exploring whether weakly supervised multitask learning can improve spam detection in cross-domain news settings. Specifically, it investigates whether auxiliary tasks such as category prediction and source prediction can enhance spam classification when trained with weak labels.

1.2 Research aim and objectives

Aim

This dissertation aims to develop and evaluate a weakly supervised multitask deep learning model for cross-domain spam news detection, using auxiliary tasks such as category and source prediction to enhance classification performance.

Objectives

To achieve this aim, the dissertation is structured around the following objectives:

- To review the literature on weak supervision, multitask learning, and spam/fake news detection in NLP.
- To design and implement a weak labelling framework for generating spam labels in a cross-domain news dataset.
- To develop baseline models for spam detection, including TF-IDF + Logistic Regression and single-task transformer classifiers.
- To design and implement multitask learning models incorporating auxiliary category and source prediction.
- To evaluate all models using standard classification metrics and compare across baselines and multitask approaches.
- To analyse results, identify strengths and limitations, and reflect on implications for weakly supervised multitask spam detection.

1.3 Research approach

This dissertation adopts a quantitative experimental research approach. It designs, trains, and evaluates competing models to explore how well weakly supervised multitask learning works for spam news detection. The research unfolds in stages, following a step-by-step design process. It begins with a baseline model, progresses to advanced transformer architectures, and ultimately extends to multitask models that include additional prediction tasks.

The dataset is obtained from open-source news corpora. It includes article text, categories, sources, and metadata such as publication dates. Because there are no manually annotated spam labels, the dataset is enhanced using a weak supervision framework. A series of heuristic labelling functions is applied to create noisy labels indicating whether an article is likely to be spam. These functions include detecting clickbait headlines, identifying excessive punctuation and capitalisation, flagging rare or unreliable sources, and noting discrepancies between titles and article content. To reduce noise, each label gets a confidence score that reflects agreement among the labelling functions. These weak labels supply the training signal for the later machine learning models.

The experimental pipeline consists of three major phases. First, traditional machine learning baselines are implemented, specifically TF-IDF features combined with Logistic Regression, to establish a reference point. Second, a single-task transformer-based model is trained to predict spam labels only, providing modern deep learning baselines.

Third, multitask learning architectures are introduced, in which a shared encoder feeds multiple output heads: one for spam classification (the primary task), and auxiliary heads for predicting article category and news source. By comparing single-task and multitask configurations, the study assesses whether auxiliary tasks enhance the quality of representations and, consequently, improve the performance of spam detection.

Model performance is carefully assessed on validation and test sets using standard classification metrics: accuracy, precision, recall, F1-score, and ROC-AUC. Confusion matrices and per-category analyses provide more insights into model behaviour. The study compares results across experiments to evaluate the effects of weak supervision, multitask learning, and auxiliary task design.

Since the research is entirely computational and depends on publicly available datasets, it does not involve human participants. Ethical considerations focus only on using data responsibly, following dataset licensing terms, and being transparent about the limitations of weak supervision.

1.4 Dissertation outline

The remainder of the dissertation is organised as follows:

- Chapter 2 - Literature Review: Surveys relevant work on weak supervision, multitask learning, and spam/fake news detection, identifying gaps this research addresses.
- Chapter 3 - Research Approach (Methodology): Details the dataset preparation, weak labelling functions, model architectures, and experimental setup.
- Chapter 4 - Data Analysis: Presents the experimental results, including baseline and multitask model performance, with visualisations and confusion matrices.
- Chapter 5 - Discussion: Interprets the findings, compares with prior work, analyses strengths and limitations, and discusses implications of weakly supervised multitask learning.
- Chapter 6 - Conclusion: Summarises contributions, highlights key insights, and outlines directions for future research.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter provides a structured review of the research related to spam and fake news detection with a particular focus on weak supervision, multitask learning (MTL), and cross-domain detection. The review begins by examining weak supervision techniques that replace manual labels with heuristic or social signals. It then explores multitask learning approaches that use auxiliary tasks to boost performance in misinformation and spam detection. Finally, it addresses cross-domain challenges and the role of auxiliary tasks in enhancing robustness and generalization. The review concludes with a critical synthesis, a comparative table and a summary that highlights research gaps and places this dissertation in the context of the broader body of work.

2.2 Weak supervision for fake/spam detection

Shu, Mahudeswaran and Liu (2019) introduced the concept of weak social supervision to address the lack of labelled data in fake news detection. Their method utilised social signals, such as user posting behaviour, sharing patterns, and comment activity, as weak labels. Using social media datasets like Twitter, they trained classifiers that combined both content features and weakly supervised social features. Results demonstrated that these models outperformed content-only baselines when gold labels were sparse, highlighting the potential of noisy but scalable social signals. However, relying on user behaviour data limits its applicability in settings where metadata is absent. This study is directly relevant to this dissertation because it supports the idea that noisy signals can be predictive, which relates to weak labelling functions (clickbait detection, punctuation heuristics, source rarity) designed in this project.

Shu et al. (2020) extended this work by proposing a multi-source weak supervision framework. Instead of relying on a single weak signal, they aggregated multiple sources, such as textual features, user metadata, and engagement patterns, using a meta-learning fusion model. When tested on multi-domain datasets, this method provided earlier and more accurate detection compared to single-source methods, even without social signals during inference. The significance lies in showing that aggregating diverse weak signals makes the system more reliable. This directly supports this dissertation's strategy of combining multiple labelling functions and computing confidence scores to weight samples during model training.

Wang et al. (2020) proposed WeFEND, a reinforcement learning framework that refined weakly labelled data. The system simulated user reports as noisy labels and applied reinforcement learning to reward labellers whose data improved validation performance. When tested on news datasets with user feedback, WeFEND consistently improved label quality and detection accuracy compared to static heuristics. The key insight is that not all weak labels are equally useful. For this dissertation, this motivates the use of confidence weighting so that more reliable labelling functions have a greater impact during training.

Özgöbek, Bayrak and Aydin (2022) focused on content-only weak supervision. They applied simple heuristics such as identifying clickbait expressions, stylometric features, and lexical anomalies. These helped them create weak labels for training traditional ML and lightweight neural models. Despite lacking external metadata, their system outperformed naive supervised baselines on news corpora. This shows that even

minimal weak supervision yields improvements. However, the limitation is that heuristic coverage is low and may not capture subtler forms of manipulation. Nevertheless, this aligns closely with the design of this dissertation, which applies textual and metadata heuristics to generate spam labels.

Akdag and Cicekli (2024) proposed WeSTeC, a framework for emerging topics where labelled data is scarce. They used adaptive weak labelling functions that change as topics evolve. They trained RoBERTa classifiers that were updated with the new labels. Experiments on news about emergent events such as pandemics showed that adaptive weak labelling functions maintained detection accuracy, while static functions degraded over time. This demonstrates the value of dynamic weak supervision, as well as the cost associated with higher computational requirements. For this dissertation, WeSTeC inspires future work to adapt labelling functions as domains change.

Xie et al. (2022) looked into the issue of label noise in weak supervision. They proposed a mean-teacher framework that trains a student model to match a teacher. The teacher updates as an exponential moving average of past weights, with consistency regularisation applied under input perturbations. When tested on noisy fake news corpora, this approach achieved more stable results and improved F1-scores compared to conventional training. For this dissertation, the relevance lies in showing that weakly labelled data must be handled carefully. The approach of applying confidence-weighted loss functions aligns with this principle of accounting for label quality.

The reviewed literature indicates that weak supervision can replace expensive manual annotation. However, the effectiveness depends on combining multiple weak signals, assigning weights to labels based on confidence or quality and ensuring robustness against noise. This dissertation follows best practices by implementing multiple heuristic labelling functions, computing spam confidence scores, and applying these weights during model training.

2.3 Multitask learning approaches for misinformation and spam

Kumari et al. (2021) demonstrated the use of multitask learning for misinformation detection by combining three tasks: novelty detection, emotion recognition, and misinformation classification. Using transformer-based encoders on news corpora, they showed that multitask learning improved misinformation detection performance by approximately 6% compared to single-task baselines. The auxiliary tasks provided complementary signals. Novelty indicated unusual content, while emotion revealed manipulative writing styles. This study confirms that auxiliary tasks aligned with the main objective can boost performance. This finding motivates the inclusion of category and source prediction alongside spam detection in this dissertation.

Spinde, Vassiliou and Schenck (2022) applied multitask learning to media bias detection. They trained transformer models with a shared encoder and multiple heads to predict different aspects of bias across six datasets. Their results highlighted that MTL produced stronger macro-F1 scores compared to training on individual datasets separately, as the shared encoder captured higher-level discourse features. The main takeaway is that multitask learning supports cross-dataset robustness. This dissertation adapts their design by training a shared encoder with heads for spam, category, and source prediction, aiming to improve generalisation across news categories.

Melleng, Aouedi and Ezzikouri (2023) explored multitask ensemble learning for fake review detection. Their model predicted both review authenticity and its helpfulness. The rationale was that helpful reviews often share linguistic and stylistic traits with legitimate ones. Results showed improved robustness and higher accuracy compared to single-task models. This supports the use of quality-related auxiliary tasks, similar to this dissertation's inclusion of category and source prediction, which injects contextual information about article style and domain.

Chelliah, Manikandan and Subramanian (2021) studied email spam detection through multitask learning. Their model incorporated auxiliary tasks such as term clustering and structural feature learning, alongside the main task of classifying spam. When tested on standard spam datasets, the multitask model achieved higher recall for spam messages, which is crucial for reducing false negatives. This provides direct proof that MTL is effective for spam classification, specifically in the email domain, supporting the decision to apply MTL for news spam detection.

Liu, Qiu and Huang (2017) proposed adversarial multitask learning for text classification. Their model separated task-invariant and task-specific features, reducing the risk of negative transfer when tasks conflicted. Evaluations across multiple NLP datasets showed improvements in both accuracy and generalisation. The key point here is that multitask learning isn't always helpful. It's important to choose auxiliary tasks wisely. This insight is valuable for interpreting the results in this dissertation, particularly where source prediction may help or hinder spam detection, depending on the task alignment.

The reviewed multitask learning studies show that auxiliary tasks improve performance when they are predictively related to the primary task. They also caution that irrelevant auxiliaries can introduce negative transfer. This dissertation carefully selects categories and sources as auxiliary tasks, as both strongly correlate with news spam patterns.

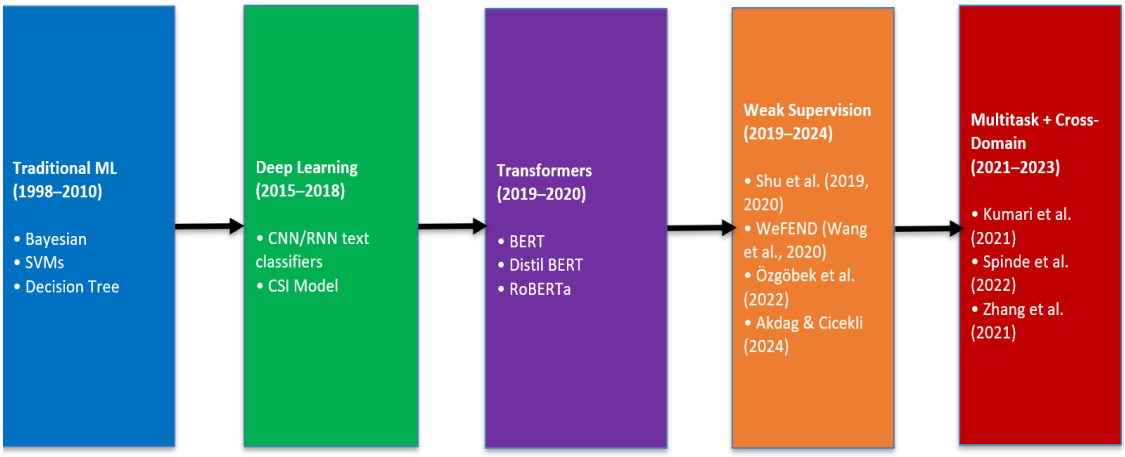


Figure 2.1: Evolution of Approaches to Fake/Spam Detection

2.4 Cross-domain challenges and the role of auxiliary tasks

Zhang et al. (2021) proposed domain-adaptive fake news detection by transferring knowledge across datasets using adversarial training. Their method reduced performance drops when applying models to unseen domains. Experiments on cross-dataset benchmarks demonstrated significant gains over naive fine-tuning. This highlights the challenge of domain shift, where models trained on one category perform poorly on others. For this dissertation, It justifies the use of a category head as a lightweight proxy for domain adaptation.

Zhang et al. (2019) developed a multimodal event memory network that combined text with knowledge bases and multimodal features to detect fake news. By capturing event-level context, their system achieved better detection accuracy than text-only baselines. The implication is that contextual signals are important for detecting misinformation. While this dissertation does not include multimodal data, the use of metadata tokens such as source and category reflects a similar attempt to inject contextual signals.

Dong et al. (2020) examined dual emotion features, capturing both the publisher's emotion in article content and the commenter's emotion in reader responses. They found that discrepancies between the two correlated strongly with the presence of misinformation. Results showed improved F1 over text-only baselines. This demonstrates that auxiliary emotion cues can enhance performance, suggesting future work for this dissertation to explore emotion as an additional auxiliary task.

Shu, Wang and Liu (2020) argued that fake news detection should go beyond content to include social context, such as propagation patterns and user credibility. Their experiments showed that models combining both content and social features consistently outperformed content-only systems. This underscores the importance of context beyond text. While this dissertation deliberately focuses on text and metadata, the results of *Shu et al. (2020)* point out a limitation and inspire future research directions.

Wu et al. (2020) proposed learning differential features for social media mining to enhance representation quality. By focusing on informative features, they improved generalisation across multiple social media tasks. This aligns with this dissertation’s use of confidence-weighted training and multi-head supervision as mechanisms to guide models toward informative representations.

The literature on cross-domain and auxiliary tasks confirms that domain adaptation is essential for robust performance, metadata or auxiliary features provide important context, and generalisation can be improved through representation shaping. This dissertation addresses these issues by incorporating category and source prediction as auxiliary tasks, evaluating whether they improve cross-domain spam detection.

2.5 Critical synthesis

From the reviewed literature, three critical insights emerge.

- Weak supervision is a practical and scalable solution to label scarcity, but its success depends on multi-signal aggregation and noise management.
- Multitask learning offers consistent performance gains when auxiliary tasks are predictively aligned, though negative transfer remains a risk.

- Domain adaptation and auxiliary signals such as category and source metadata are essential for cross-domain robustness.

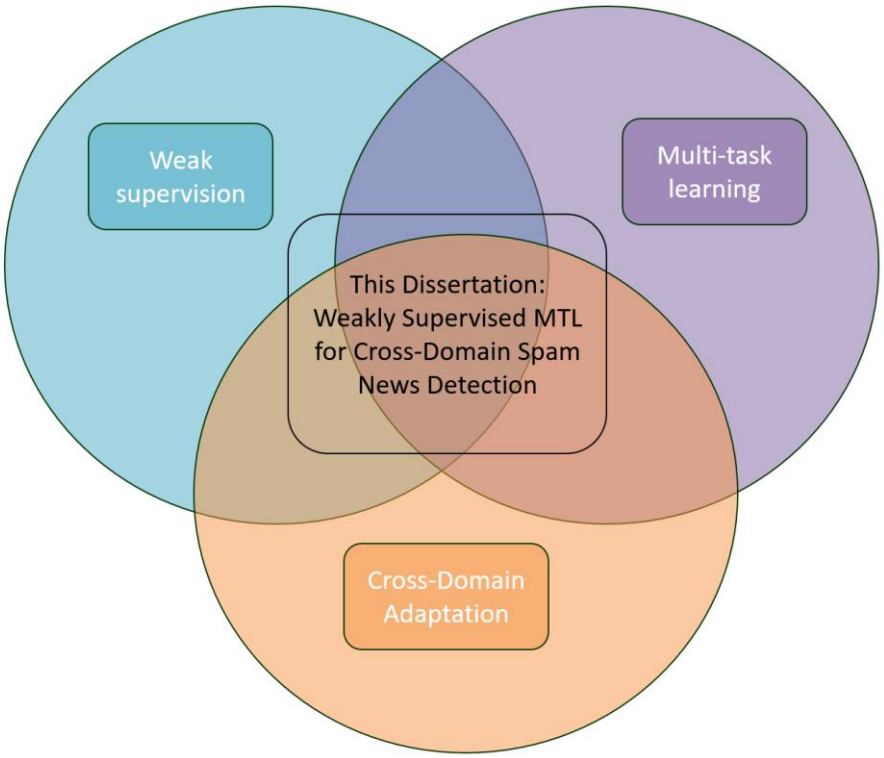


Figure 2.2: Conceptual positioning of the dissertation within existing research streams.

This dissertation is positioned at the intersection of these findings. It combines weak supervision and multi-task learning to detect spam in various news categories. By using multi-function weak labelling, confidence weighting, and shared-encoder multi-task architecture, it provides evidence for a relatively underexplored area of research.

2.6 Comparative summary

STUDY	APPROACH	DATA TYPE	KEY FINDINGS	LIMITATION	RELEVANCE HERE
Shu et al. (2019)	Weak social supervision	Twitter	Social signals bootstrap labels	Domain-specific	Validates the weak labels concept
Shu et al. (2020)	Multi-source weak supervision	News/social	Early detection without social inference	Complexity	Justifies multi-LF design
Wang et al. (2020)	RL + weak reports (WeFEND)	News	Higher-quality weak labels	Heavy compute	Inspires confidence weighting

Özgöbek et al. (2022)	Heuristics + ML	News	Content-only weak labels work	Low coverage	Supports heuristic LF design
Akdag & Cicekli (2024)	Adaptive weak labels	Emerging topics	Dynamic adaptation improves accuracy	Resource intensive	Future work direction
Xie et al. (2022)	Noise-resistant weak supervision	Fake news corpora	Stable results with noisy labels	Requires consistency training	Validates weighting approach
Kumari et al. (2021)	MTL (novelty, emotion, misinformation)	News	Aux tasks improve F1	Needs rich labels	Supports MTL design
Spinde et al. (2022)	Transformer MTL for bias	Multi-domain	Shared encoder boosts robustness	Task-specific	Supports shared encoder
Melleng et al. (2023)	MTL ensemble for reviews	Reviews	Helpfulness aux boosts robustness	Domain-specific	Analogous to category/source
Chelliah et al. (2021)	MTL for email spam	Emails	Improved recall	Email-specific	Confirms MTL works for spam
Liu et al. (2017)	Adversarial MTL	NLP tasks	Reduces negative transfer	Complexity	Explains task conflicts
Zhang et al. (2021)	Domain-adaptive fake news	News	Transferable features	Needs adaptation data	Justifies the category head
Zhang et al. (2019)	Multimodal event memory	News+multimodal	Context boosts detection	Needs extra modalities	Metadata tokens as proxy
Dong et al. (2020)	Dual emotion features	News+comments	Emotion discrepancy predictive	Needs reader data	Future auxiliary signal
Shu et al. (2020, WSDM)	Content + social context	News+social	Social context boosts performance	Not always available	Acknowledges limitations

Table 2.1: Comparison of weak supervision, multitask learning, and cross-domain strategies in fake/spam detection

2.7 Summary

This chapter reviewed literature on weak supervision, multitask learning, and cross-domain detection for spam and fake news. Weak supervision is essential for overcoming label scarcity, but it requires aggregation and effective noise handling. Multitask learning enhances performance with aligned auxiliary tasks, but it also risks negative transfer. Cross-domain robustness benefits from auxiliary metadata and domain adaptation techniques. The gap identified is the limited exploration of weakly supervised multitask learning for cross-domain spam detection. This dissertation addresses that gap by developing and evaluating shared-encoder multitask models with spam as the primary task and category and source as auxiliary tasks. The next chapter describes the research methodology (CRISP-DM), dataset processing, weak labelling framework, model design, and evaluation protocol.

CHAPTER 3: RESEARCH APPROACH

3.1 Introduction

This chapter outlines the research methodology adopted to address the aim of this dissertation: *To develop and evaluate a weakly supervised multitask deep learning model for cross-domain spam news detection*. Building on the insights from the literature review, the study follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework, which provides a structured yet iterative approach suitable for experimental data science research.

The chapter begins by explaining the rationale for selecting CRISP-DM over other established methodologies. It then details the implementation of each CRISP-DM phase in the context of this project, including business understanding, data understanding, data preparation, modelling, evaluation, and deployment. Ethical considerations and data access arrangements are also discussed. The chapter concludes with a summary that highlights the alignment between methodology and research objectives.

3.2 Choice of methodology: CRISP-DM

3.2.1 Overview

CRISP-DM is a six-phase process model widely used in data mining and machine learning projects. The phases include: (i) business understanding, (ii) data understanding, (iii) data preparation, (iv) modelling, (v) evaluation, and (vi) deployment. This methodology is iterative, allowing researchers to revisit earlier phases based on new insights or unexpected challenges. This flexibility makes CRISP-DM highly suitable for exploratory research projects such as this dissertation, which involve iterative weak labelling and experimental model development.

3.2.2 Comparison with alternatives

Several alternative research methodologies exist in the data mining and data science literature, each with strengths and weaknesses. While they provide valuable structures, none align as closely with the requirements of this dissertation as CRISP-DM.

- *KDD (Knowledge Discovery in Databases)*

  KDD is one of the earliest formal methods. It includes steps like selection, preprocessing, transformation, data mining, and interpretation. The method emphasises discovering patterns and is particularly effective for exploring large databases. However, KDD has faced criticism for being somewhat linear in its execution and for having few opportunities for feedback. In situations like weak supervision, where labelling functions often require improvement after initial model results, this lack of iteration can be limiting. Additionally, KDD focuses more on the technical aspects of mining patterns rather than clearly incorporating business or research goals into the process.

- *SEMMA (Sample, Explore, Modify, Model, Assess):*

  SEMMA was developed by SAS Institute and is closely linked to the SAS Enterprise Miner software. It provides a structured approach that begins with

sampling data, followed by exploration, modification, modelling, and assessment. Its main strength is a practical workflow for data preparation and modelling, especially for applied analytics. However, SEMMA has two main limitations in the context of this dissertation. First, it is predominantly tool-driven, often associated with commercial software environments rather than open-source, research-oriented pipelines. Second, SEMMA focuses mainly on the modelling and analysis phase, providing little explicit attention to problem framing, evaluation of success criteria, or ethical considerations. For academic research, where transparency and broader framing are critical, this narrower focus is a limitation.

- *GABDO (Goal–Analysis–Business understanding–Data preparation–Operation):*

  GABDO is a newer methodology that is not widely used. It begins with establishing the project's goals and analysing requirements, followed by business understanding, data preparation, and implementing the process. Although it emphasises goal-setting and business analysis, there is a lack of documentation in academic literature, resulting in limited practical application. Its lack of standard application results in fewer references, tools, and community practices that could help with its implementation. For a dissertation project where methodological rigour and comparability with prior work are essential, GABDO does not provide the same level of legitimacy or support as CRISP-DM.

In summary, while KDD, SEMMA, and GABDO each provide valuable perspectives on managing data-driven projects, CRISP-DM remains the most appropriate for this dissertation. Its comprehensive lifecycle, iterative design, and broad adoption in both academic and industrial contexts make it a robust choice for guiding a project that combines experimental weak supervision, deep learning modelling, and comparative evaluation.

3.2.3 Justification for CRISP-DM

CRISP-DM was selected as the guiding methodology for this dissertation because it provides comprehensive, end-to-end coverage of the data science research lifecycle, encompassing the stages of problem formulation, data handling, model building, evaluation, and final deployment. Unlike narrower methodologies, CRISP-DM explicitly integrates both the conceptual framing of the problem and the technical implementation of solutions. This makes it compatible with academic projects that must balance theoretical contribution with empirical experimentation.

A key reason for choosing CRISP-DM is its iterative design. Weak supervision and multitask deep learning are inherently experimental processes: initial heuristic labelling functions may need refinement once preliminary evaluation reveals their limitations, and model architectures often undergo multiple revisions to balance task losses or address negative transfer. CRISP-DM supports such back-and-forth refinement without breaking methodological consistency. In contrast, more linear methodologies such as KDD provide fewer mechanisms for revisiting earlier stages.

CRISP-DM also places strong emphasis on the data preparation phase, which is particularly critical in this project. Because the dataset did not include ground-truth spam labels, significant effort was required to design and apply weak labelling functions, aggregate them into usable labels, and compute confidence scores. CRISP-

DM explicitly recognises data preparation as a standalone phase, rather than treating it as a minor preliminary step, thereby validating the centrality of this process in the dissertation.

Another justification lies in the evaluation phase. The dissertation involves not only testing models on validation and test splits but also systematically comparing baselines against multitask learning approaches, analysing per-category spam recall, and exploring cross-domain robustness. CRISP-DM's evaluation stage provides a formal structure for this comparative analysis, ensuring that results are measured against predefined success criteria and not interpreted randomly.

Finally, CRISP-DM is widely used in both academic research and industry practice. This makes it a credible and defensible methodological choice. Its widespread recognition ensures that examiners easily understand the methodological framing of this dissertation and aligns with best practice in the field of data science. In summary, CRISP-DM was chosen not only because of its completeness and flexibility but also because its structure directly supports the requirements of weak supervision, multitask modelling, and rigorous evaluation in cross-domain spam news detection.

3.3 CRISP-DM phases applied to this dissertation

The research methodology for this dissertation is based on the CRISP-DM (Cross Industry Standard Process for Data Mining) framework, which is one of the most widely adopted standards in data science. CRISP-DM outlines six iterative phases: business understanding, data understanding, data preparation, modelling, evaluation, and deployment. These phases ensure that projects progress systematically from problem definition to actionable outcomes.

While CRISP-DM provided the conceptual foundation for this study, its phases were adapted into a customised three-part pipeline tailored to the dissertation's aims. This was necessary to avoid redundancy and to reflect the specific challenges of weak supervision better and multitask learning. The adapted pipeline consists of:

- Weak labelling and dataset preparation
- Baseline and multitask model development
- Evaluation and comparative analysis

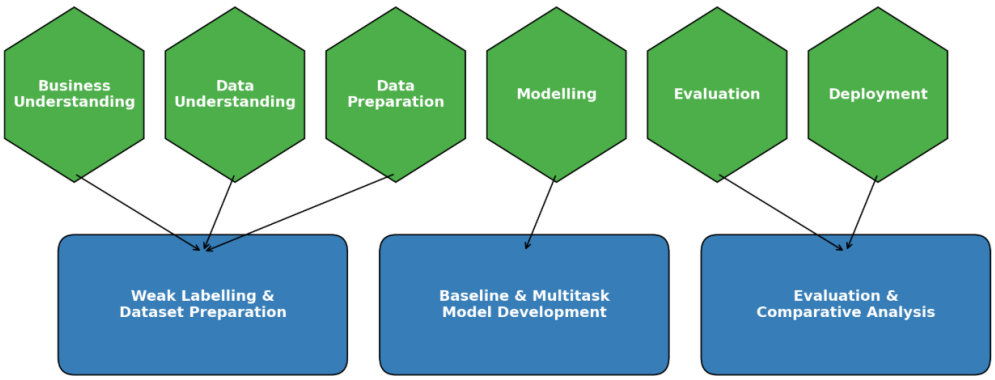


Figure 3.1: Mapping CRISP-DM phases to adapted dissertation phases

This restructuring adheres to the CRISP-DM methodology while presenting the research process in a manner that directly aligns with the dissertation's objectives. The following subsections provide a detailed description of each part of the adapted pipeline.

3.3.1 Weak labelling and dataset preparation

In the absence of reliable spam labels, the first stage focused on creating a usable dataset through weak supervision. Heuristic labelling functions were designed to capture common spam characteristics, such as clickbait patterns, excessive punctuation, source rarity, duplicate titles, and mismatches between headlines and content. These functions produced noisy labels, which were aggregated using confidence voting. Confidence scores were computed to reflect the reliability of each weak label, enabling more robust training later in the process.

Alongside labelling, the dataset was pre-processed by cleaning text (removing HTML tags, links, and noise), normalising case and punctuation, and splitting into training, validation, and test sets with stratified sampling to preserve category distributions. This phase established the foundation dataset on which all subsequent models were trained.

3.3.2 Baseline and multitask model development

The second stage of the research translated the prepared dataset into experimental models through a tiered modelling strategy designed to enable step-by-step comparison. At the simplest level, a traditional TF-IDF combined with a Logistic Regression classifier was implemented to provide a fast and interpretable benchmark against which more sophisticated approaches could be measured. Building on this, transformer-based single-task spam classifiers were developed using RoBERTa-base as the encoder, fine-tuned with binary cross-entropy loss and enhanced through class weighting to counter class imbalance, with optional confidence weighting to account for noisy weak labels.

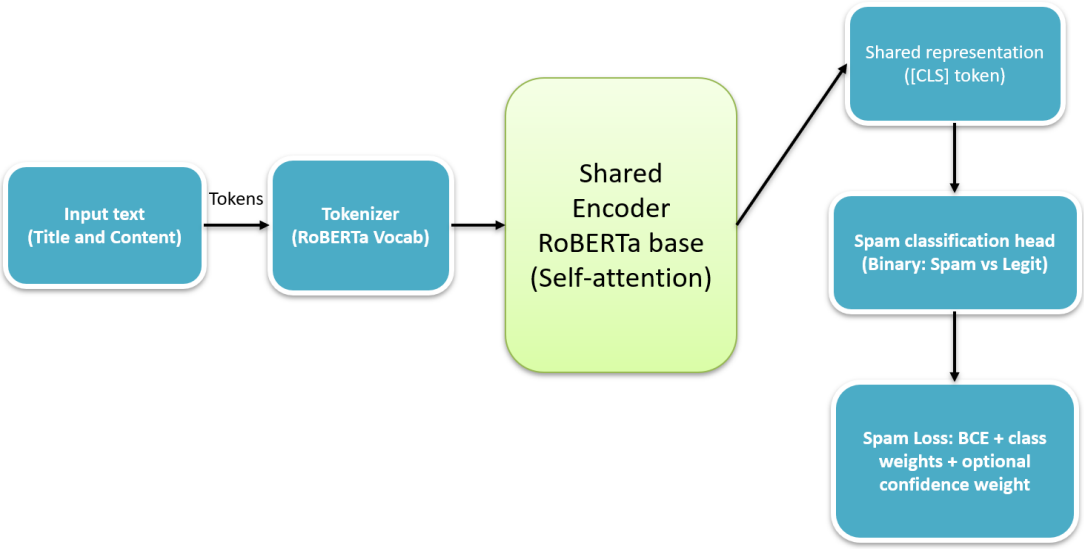


Figure 3.2: Single-task transformer model for spam detection

Finally, the core contribution of the dissertation was the design of multitask learning models, where a shared RoBERTa encoder was extended with multiple task-specific heads to jointly predict spam detection, category classification, and source identification. The three tasks were optimised using a weighted sum of loss functions, with spam detection prioritised (α = 1.3) and auxiliary tasks for category (β = 1.0) and source (γ = 0.5) providing additional signals to guide representation learning. By progressing systematically from traditional baselines to advanced multitask architectures, this modelling stage ensured that any observed performance improvements could be attributed specifically to the integration of weak supervision and multitask learning, rather than merely the adoption of larger neural models.

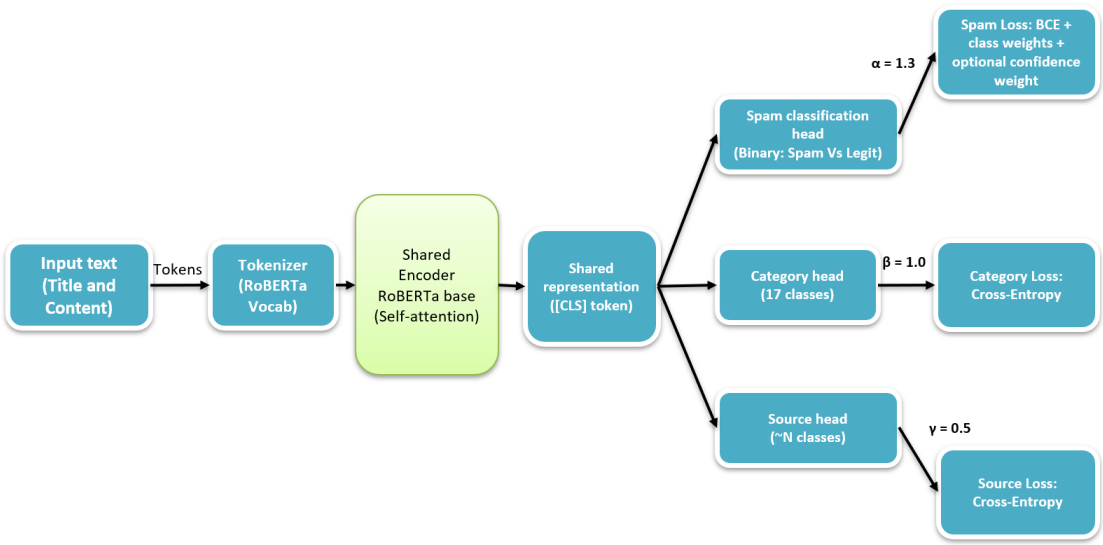


Figure 3.3: Multi-task transformer model (spam + Category + Source)

3.3.3 Evaluation and comparative analyses

The final stage of the methodology concentrated on evaluating how well the models performed under weak supervision and on determining whether the inclusion of auxiliary tasks enhanced robustness across domains. To provide a reliable and comprehensive picture, multiple metrics were employed. Spam detection was evaluated using accuracy, precision, recall, F1-score, and ROC-AUC, with F1-score and ROC-AUC serving as the primary indicators due to the class imbalance between legitimate and spam articles. The auxiliary tasks of category and source prediction were evaluated separately using accuracy and Macro-F1, with the latter chosen to ensure that performance was not dominated by majority classes. Cross-domain robustness was assessed through per-category spam recall, which revealed whether detection accuracy was stable across topical domains such as politics, health, and society. Confusion matrices were also produced for validation and test sets to provide visual diagnostics of the distribution of predictions and to highlight systematic biases, such as a tendency to overpredict the legitimate class.

Comparisons were then drawn across different modelling approaches to isolate the contribution of each methodological choice. Traditional TF-IDF with Logistic Regression was contrasted with transformer-based baselines to quantify the gains

from contextual embeddings. Single-task classifiers were compared against multitask architectures to evaluate the value of auxiliary supervision. Additional experiments examined the effect of applying or omitting confidence weighting to weak labels, allowing an assessment of how weighting schemes influenced robustness. Finally, different multitask configurations, such as spam with category versus spam with category and source, were tested to determine whether source prediction added meaningful auxiliary information or introduced negative transfer. Together, this multi-layered evaluation framework ensured that the analysis went beyond raw accuracy, providing a clear understanding of how weak supervision and multitask learning shaped the effectiveness of spam detection models.

3.4 Ethical considerations

Ethical issues in this dissertation were relatively limited, as the study involved no direct human participation. Nonetheless, several considerations were addressed to ensure responsible research practice:

- *Data privacy:* The dataset contained only article text, categories, publication sources, and dates. No personally identifiable information (PII) was included, eliminating risks to individual privacy.
- *Use of sources:* Publication sources were used solely as categorical features for modelling. No judgments were made about the credibility or trustworthiness of individual outlets; predictions were framed as spam detection rather than reputational assessment.
- *Weak labelling noise:* Recognising that weak labelling functions may misclassify legitimate articles, the methodology incorporated confidence weighting to reduce the influence of uncertain labels. This mitigated the risks of unfair bias in training data.
- *Licensing and openness:* All datasets were sourced from Zenodo, a repository for open scientific data, and were used in full compliance with licensing terms. Pre-processing and labelling scripts were designed for reproducibility, ensuring transparency of the pipeline.

By addressing these considerations, the research aligns with ethical standards for data science projects. It avoids potential risks associated with misinformation research, such as unfair stigmatisation of sources or privacy breaches.

3.5 Access to data

The dataset employed in this dissertation was obtained from the Zenodo repository, a widely recognised open-access platform for academic and scientific datasets. The need for transparency, reproducibility, and legitimacy in data-driven research guided the choice of Zenodo. Open repositories not only allow unrestricted use for academic purposes but also ensure that other researchers can replicate or extend the work by accessing the same source material.

The raw dataset comprised an extensive collection of news articles, each accompanied by rich metadata, including title, content, category labels, source of publication, and publication date (Petukhova and Fachada, 2022). This combination of textual and metadata features made the dataset particularly well-suited for the goals of this dissertation. Specifically, the presence of categories enabled the exploration of cross-domain challenges, as models could be evaluated across diverse topical domains,

such as politics, health, sports, and society. The inclusion of sources allowed the investigation of source prediction as a potential auxiliary task, capturing stylistic and editorial patterns. The timestamps facilitated analysis of temporal patterns, including the “burstiness” heuristic used in weak labelling.

To ensure the dataset was ready for experimental use, a comprehensive preprocessing pipeline was designed and implemented. This included cleaning text, applying weak labelling functions to generate spam labels, and computing confidence scores to reflect label reliability. Finally, the data was split into training, validation, and test sets using stratified sampling to preserve category balance and avoid bias in evaluation. These processed splits were stored separately to ensure reproducibility and consistency across experiments.

Despite these strengths, the dataset also posed notable limitations. The absence of ground-truth spam labels required reliance on weak supervision, which introduces noise and potential misclassification. The dataset exhibited imbalanced class distributions, both at the spam/non-spam level and across categories and sources. For example, categories such as politics were heavily represented, while domains like weather were under-represented. Similarly, many sources appeared only once or twice, resulting in long-tail distributions that are challenging to model effectively.

Thus, while imperfect, the dataset provided a realistic and challenging environment in which to evaluate weakly supervised multitask learning approaches for spam detection.

3.6 Chapter summary

This chapter outlines the research approach adopted to address the dissertation's aim. The CRISP-DM was selected as the methodological backbone because of its comprehensive coverage of the data science lifecycle and its iterative nature, which aligns with the experimental design of this study. In summary, this chapter demonstrated that the research was based on a solid and widely accepted methodology, tailored to the specific challenges of weakly supervised spam detection. By systematically applying the CRISP-DM methodology, the project ensured methodological consistency, clarity of evaluation, and responsible data usage. The next chapter, Data Analysis, presents the observed outputs of the research process. It demonstrates the application of the CRISP-DM framework in practice by showing dataset preparation, weak labelling outcomes, baseline model training, and multitask experiments. The chapter focuses on reporting results systematically and transparently, including performance metrics, confusion matrices, and category-level spam recall.

CHAPTER 4: DATA ANALYSIS

This chapter presents the data analysis carried out in this dissertation, demonstrating the application of the CRISP-DM methodology to the problem of weakly supervised multitask spam news detection. It reports the outputs of each phase of the analysis pipeline, from understanding and preparing the dataset to training baseline and multitask models and reporting evaluation results.

This chapter not only describes the steps undertaken but also justifies the choice of analytical techniques and model configurations, drawing on relevant literature to ensure methodological rigor. By following the framework established in Chapter 3, this chapter provides evidence of proficiency in data analysis, ensuring that design decisions are both theoretically and practically grounded.

4.1 Business understanding

The business understanding phase of CRISP-DM translates into research framing in an academic context. The motivation for this dissertation arises from the increasing prevalence of low-quality or deceptive content on online news platforms. Spam news, often characterised by clickbait headlines, exaggerated claims, or duplicated and recycled articles, undermines the credibility of information ecosystems. Tackling this problem is both technically challenging and socially significant, given the role of news media in shaping public opinion.

The overarching research question guiding this dissertation is therefore:

Can weakly supervised multitask learning improve cross-domain spam news detection compared to single-task and baseline models?

This question reflects three critical challenges identified in the literature review:

- The scarcity of high-quality labelled data,
- The difficulty of capturing the multifaceted nature of spam signals
- The problem of cross-domain generalisation, whereby models trained on one category of news often fail to transfer effectively to others.

To address this question, the following objectives were formulated.

- First, *to generate weak labels for spam detection* using multiple heuristic labelling functions (LFs), including clickbait pattern detection, punctuation analysis, and rarity of publication sources. This directly responds to the challenge of label scarcity, a problem widely recognised in misinformation detection research (Shu, Mahudeswaran and Liu, 2019).

- Second, *to develop baseline models*, including TF-IDF with Logistic Regression as a traditional benchmark and transformer-based single-task spam classifiers such as RoBERTa as modern baselines. These establish reference points for evaluating the added value of multitask approaches.

- Third, *to design multitask models that incorporate auxiliary tasks* such as category and source prediction, leveraging prior evidence that metadata

provides useful contextual signals for detecting deceptive content (Spinde et al., 2022; Zhang et al., 2021).

- Fourth, *to evaluate the performance of all models* across validation and test datasets using established metrics such as accuracy, precision, recall, F1-score, and ROC-AUC, ensuring that improvements are measured rigorously. Finally, to assess cross-domain robustness by analysing spam recall across different news categories, thereby determining whether multitask learning helps mitigate domain shift.

The success criteria against which the research is evaluated combine both quantitative and qualitative dimensions. Quantitatively, the models should achieve improvements in F1-score and ROC-AUC relative to both traditional and transformer-based baselines, in line with accepted standards in spam and misinformation detection (Wang et al., 2020). Qualitatively, the models should demonstrate evidence of cross-domain generalisation, reflected in stable spam detection performance across categories such as politics, health, and sport. Additionally, the work must demonstrate that weak supervision can serve as a viable substitute for costly manual labelling, providing a scalable framework that can be applied in real-world settings such as news aggregators and social platforms.

By framing the project in this way, the business understanding phase establishes not only the academic relevance of the dissertation but also its practical significance. It connects the research objectives with broader debates in the literature, aligns success criteria with established evaluation practices, and situates the work as a contribution to ongoing efforts to mitigate the spread of spam and low-quality information in digital news ecosystems.

4.2 Data understanding

The data understanding phase involved a systematic examination of the dataset with the dual aim of identifying its opportunities and challenges for downstream modelling. The dataset used in this dissertation was sourced from the Zenodo Fake/Spam News repository, a publicly available collection of thousands of news articles enriched with metadata (Petukhova and Fachada, 2022). Each record includes the article's title, full textual content, category label (e.g., politics, health, society, sport), publication source, and publication date. The inclusion of both textual and metadata fields made this dataset particularly well-suited to the project, as prior research has emphasized that metadata, such as categories and sources, can provide contextual signals and enhance misinformation detection (Zhang et al., 2021; Spinde et al., 2022). However, the dataset lacks explicit labels for spam or legitimacy, which necessitated the design of a weak supervision framework to generate them.

Exploratory data analysis (EDA) revealed several important patterns and potential challenges.

- *Category distribution:* The dataset was found to be highly imbalanced. While the dataset spans 17 categories, politics and society accounted for a disproportionately high number of articles, whereas categories such as weather and lifestyle were sparsely represented. An imbalance of this kind is well-documented in text classification (Johnson and Khoshgoftaar, 2019), where it risks introducing bias into both training and evaluation. This observation

highlighted the need for stratified sampling during dataset splitting and justified incorporating category prediction as an auxiliary task to support generalisation.

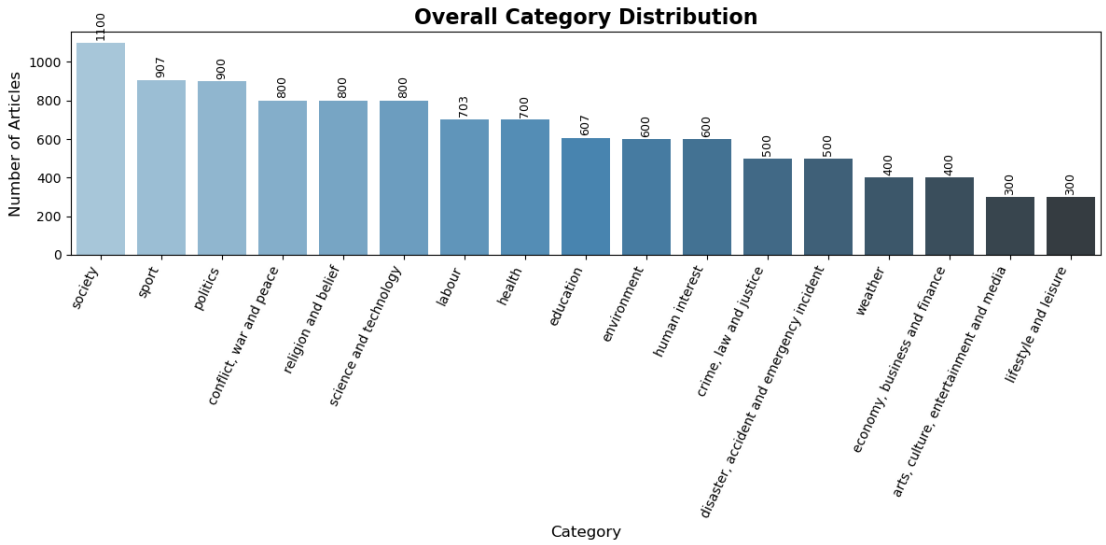


Figure 4.1: Overall Category distribution

- *Source diversity:* The dataset includes articles from hundreds of unique sources, ranging from major media outlets to less frequent contributors. Many sources appear fewer than ten times, creating a long-tail distribution. This characteristic complicates source-based modelling, since rare classes are inherently difficult to predict (Buda, Maki and Mazurowski, 2018). At the same time, the diversity of sources highlights the potential value of incorporating source prediction as a multitask objective, treating it as a noisy yet informative signal of style, structure, and credibility.

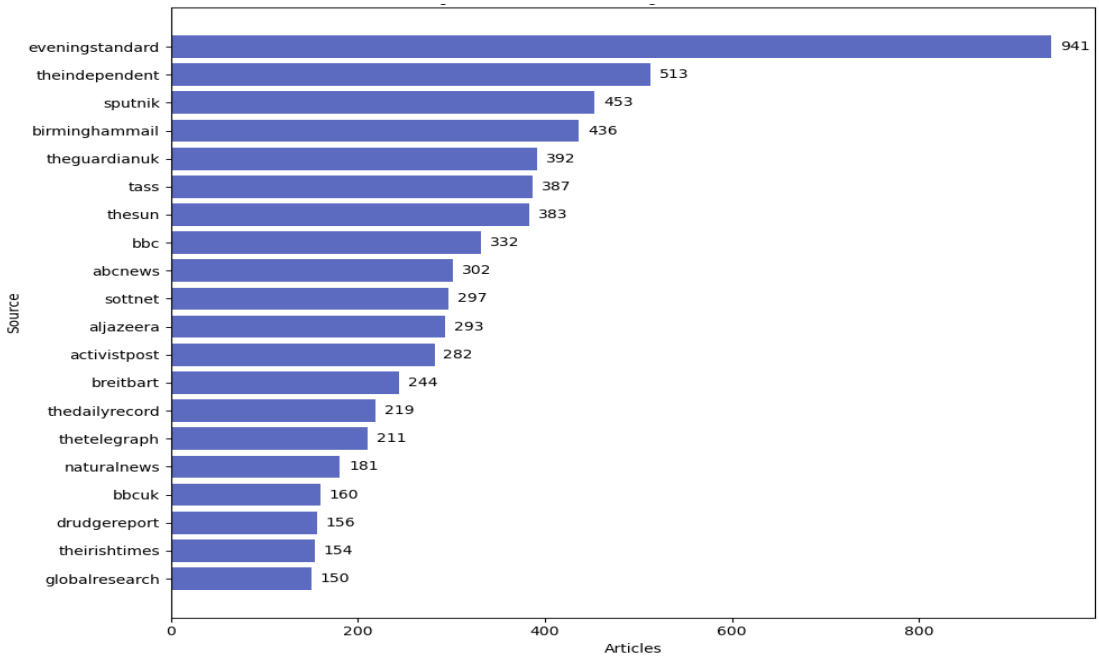


Figure 4.2: Distribution of top 20 sources based on article count

- *Textual variability:* There was marked textual variability across the corpus. Some articles were extremely short, consisting of fewer than 50 words and resembling teasers or summaries, while others exceeded 2,500 words, aligning more with long-form reports or raw data dumps. Previous work has shown that such variation is typical of news datasets (Shu et al., 2020) and that it influences classification performance. This indicated that spam detection would need to capture stylistic and structural cues, rather than relying solely on superficial features such as article length.

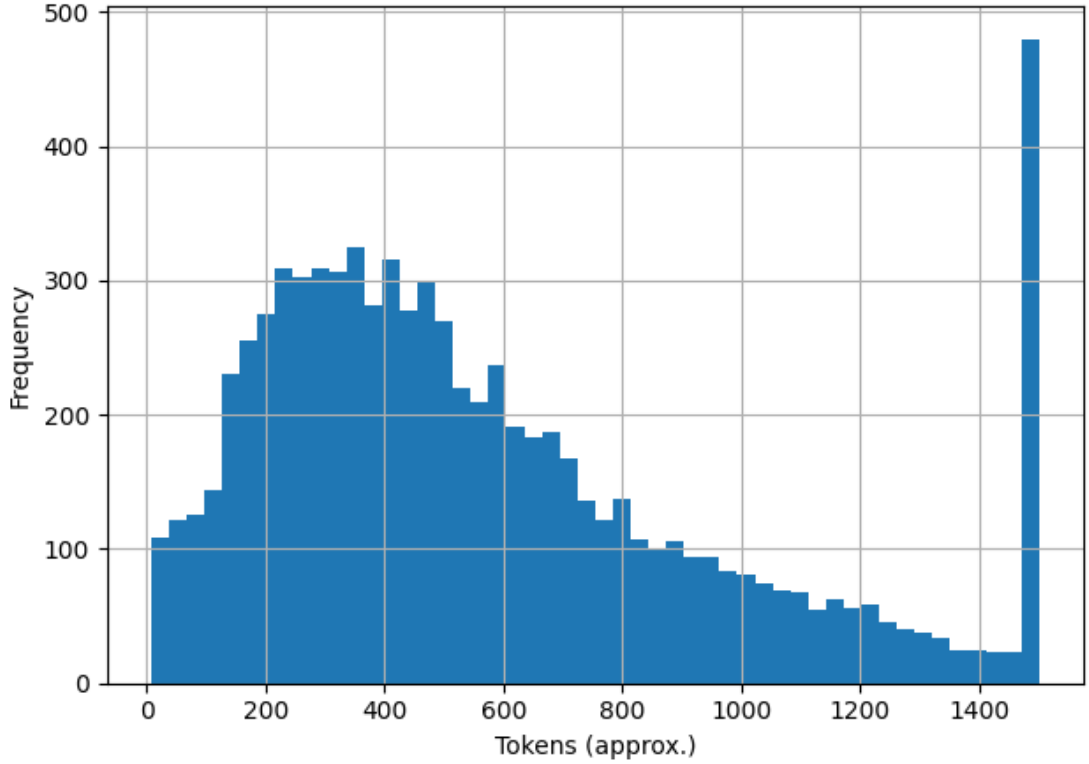


Figure 4.3: Text length distribution in training set

- *Potential noise:* EDA identified instances of duplicated or near-duplicated titles that appeared across different sources, and extreme outliers in article length were identified. These patterns are consistent with spam behaviours such as content recycling or mass distribution of identical headlines (Özgöbek et al., 2022). While such anomalies provide useful weak signals for labelling, they also risk misleading models if not carefully managed.

Taken together, these findings carried critical methodological implications.

- ➢ The absence of ground-truth spam labels required the development of heuristic labelling functions to approximate classification.
- ➢ The variability in reliability of weak labels justified the introduction of confidence scoring, ensuring that noisier samples exerted less influence during training (Xie et al., 2022).

- The imbalance in category distribution reinforced the need for stratified sampling in creating training, validation, and test splits, to preserve representativeness and avoid leakage.

In summary, the data understanding phase confirmed that while the Zenodo dataset is rich in both textual and metadata features, it also presents challenges of imbalance, diversity, and noise. These challenges directly motivated the weak supervision framework and the multitask learning design that reinforce the later stages of this dissertation.

4.3 Data preparation

Data preparation was a critical stage of this dissertation, as the dataset lacked gold-standard labels for spam or legitimacy. Instead of relying on costly manual annotation, a weak supervision pipeline was designed to generate approximate spam labels and associated confidence scores. This approach reflects best practice in spam and fake news detection research, where weak supervision has emerged as a viable strategy for overcoming label scarcity (Shu, Mahudeswaran and Liu, 2019; Akdag and Cicekli, 2024). By creating a structured data preparation process, the project ensured that models could be trained on sufficiently large data while mitigating the risks of noise and bias inherent in weak labels.

The preparation pipeline comprised four stages: text cleaning, weak labelling functions, label aggregation, and dataset splitting.

4.3.1 Text cleaning

The raw dataset contained noise in the form of embedded hyperlinks, HTML tags, irregular line breaks, duplicated characters, and excessive punctuation. To improve feature consistency, a standard preprocessing pipeline was implemented. This included removing HTML tags and URLs, converting all text to lowercase, normalising whitespace, and filtering excessive punctuation. Such cleaning steps are widely recommended in natural language processing pipelines (He and Garcia, 2009). They are essential for both the heuristic labelling functions and for downstream transformer-based models, which require consistent tokenized input. Tokenisation was handled using RoBERTa's subword vocabulary to preserve rare word fragments and stylistic cues (Liu et al., 2019).

4.3.2 Weak labelling functions

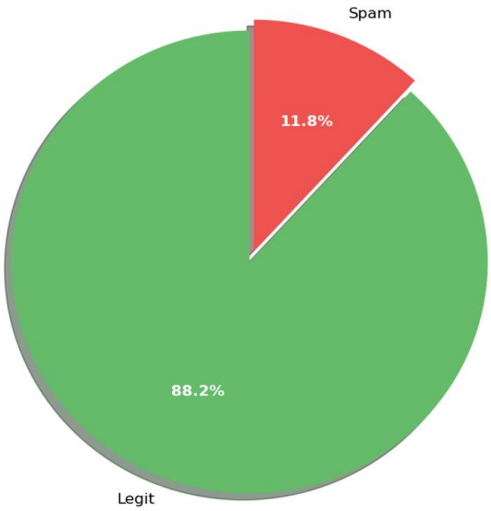
A set of heuristic labelling functions (LFs) was designed to capture patterns commonly associated with spam content. Each LF was informed by prior literature and exploratory data analysis. The functions included

- **Clickbait patterns:** Detection of exaggerated or sensational expressions such as "you won't believe", "shocking", and "what happens next" inspired by work on clickbait detection (Chen et al., 2015; Wei and Wan, 2017). Clickbait titles are widely associated with spammy or low-credibility content.
- **Excessive punctuation or capitalisation:** Headlines with multiple exclamation marks or all-caps text were flagged as suspicious, based on findings that spammy headlines often contain repeated exclamation marks or all-caps text (Potthast et al., 2018).

- **Content length extremes:** Articles with very short bodies (under ~40 words) were considered low-effort spam, while extremely long articles (over ~2500 words) were suspected of being data dumps or automated reposts.
- **Rare publication sources:** Sources with very few entries in the dataset were given higher prior probabilities of being spam, on the assumption that established media are less likely to publish spam.
- **Duplicate titles across sources:** Articles with identical titles appearing across many outlets were labelled as potentially spammy, since duplication can indicate automated distribution or low-quality aggregation.
- **Title - content semantic mismatch:** Using TF-IDF vectorisation, cosine similarity was computed between an article's title and its content. A very low similarity score suggested that the headline misrepresented the body, a common spam tactic noted in misinformation research
- **Source burstiness:** If a source published a large number of articles within a very short time window (e.g., dozens in 30 minutes), the articles were flagged as suspicious, since this behaviour often corresponds to automated spam accounts (Volkova et al., 2017).

Each LF returned either a spam vote or abstained. These heuristics were deliberately broad to capture diverse patterns of suspicious content, consistent with weak supervision systems that combine heterogeneous signals (Ratner et al., 2017).

4.3.3 Label aggregation




```
Weak labels created. Spam rate: 0.118 | abstain-as-legit count: 9627
Total news articles: 10917
Spam-labeled articles: 1290
```

Figure 4.4: Overall Spam Vs Legit distribution

Because no single LF was reliable on its own, their outputs were aggregated using majority voting, a standard technique in weak supervision frameworks (Ratner et al., 2017). If more than half of the active functions flagged an article as spam, the article was weakly labelled as spam. Additionally, a confidence score was calculated as the ratio of LFs voting spam to the total number of LFs triggered for that sample. This

allowed models to weigh samples differently during training, giving higher importance to articles with a strong consensus on spam. This weighting strategy is supported by recent research, which shows that incorporating confidence improves robustness under noisy labels (Xie et al., 2022). Articles where no functions were fired were defaulted to legitimate classification.

4.3.4 Data splitting

The dataset was partitioned into training (70%), validation (15%), and test (15%) sets. Stratified sampling was used to preserve category proportions across splits, avoiding bias towards dominant domains such as politics and society. Maintaining representative distributions is particularly important in imbalanced text classification problems (He and Garcia, 2009), and ensures that performance metrics reflect genuine cross-domain robustness.

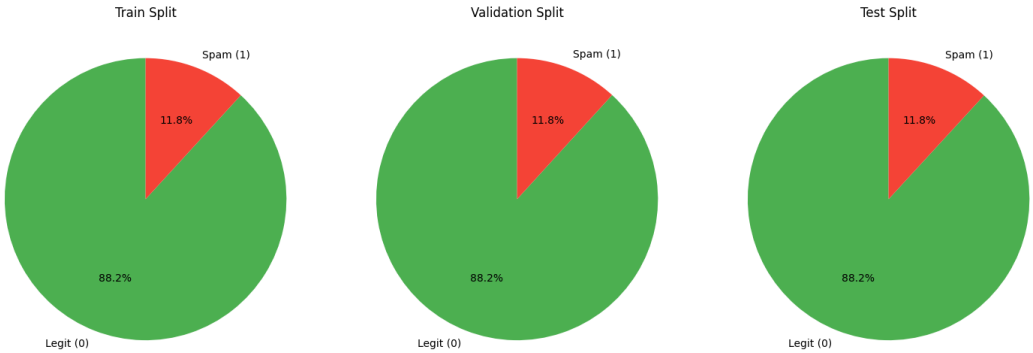


Figure 4.5: Spam Vs Legit distribution across train, validation, and test sets

The outcome of this phase was a processed dataset enriched with weak spam labels and confidence scores. This dataset served as the foundation for all subsequent experiments, ensuring that both baselines and multitask models could be trained and evaluated consistently. By combining careful preprocessing, heuristic-based labelling, confidence-weighted aggregation, and principled dataset splitting, the data preparation stage demonstrated proficiency in data analysis techniques. It aligned with established practices in weakly supervised learning.

4.4 Model development

Model development followed a tiered approach, beginning with traditional baselines and progressing to transformer-based multitask architectures. This progression reflects standard practice in empirical machine learning, where simple models are first established as benchmarks before more advanced methods are introduced (Zhang et al., 2021).

4.4.1 Baseline 1: TF-IDF with logistic regression

The first baseline used a classical machine learning pipeline. Articles were vectorised using Term Frequency - Inverse Document Frequency (TF-IDF), a sparse representation that captures the relative importance of words in documents. A Logistic Regression classifier was then trained on these features to predict spam versus legitimate news.

- *Strength:* This approach is computationally efficient, interpretable, and quick to train, making it a strong lower-bound benchmark (Delany et al., 2005). It also provides insight into whether more complex models are genuinely adding value.
- *Limitation:* TF-IDF cannot capture semantic relationships, word order, or contextual meaning. It struggles with long articles where word frequency alone fails to reflect writing style or coherence, which are critical in distinguishing spam from legitimate articles.

4.4.2 Baseline 2: Transformer-based single-task classifiers

The second baseline adopted a transformer-based single-task classifier using RoBERTa-base. Transformer models are the current state of the art in natural language processing (Devlin et al., 2019; Liu et al., 2019) and have been shown to significantly outperform traditional baselines in fake news detection tasks (Kaliyar, Goswami and Narang, 2021). RoBERTa was chosen specifically because of its robust pre-training on large corpora and proven effectiveness in text classification.

- *Input:* Article title and content were concatenated into a single sequence, truncated or padded to a maximum length of 192 tokens to meet model constraints.

- *Training:* Training was conducted for four epochs with a batch size of 8 for training and 32 for evaluation, and a learning rate of 2e-5 was used with AdamW optimisation and linear scheduling. These hyperparameters are consistent with best practices for fine-tuning transformers (Howard and Ruder, 2018; Devlin et al., 2019). Gradient clipping at 1.0 and dropout regularisation were employed to stabilise training and mitigate overfitting.

- *Loss function:* The models were fine-tuned using binary cross-entropy loss, with class weights applied to counter the imbalance between legitimate and spam articles, consistent with recommendations in imbalanced learning literature (Johnson and Khoshgoftaar, 2019). Confidence weighting was optionally applied to down-weight noisy weak labels, following approaches in weakly supervised text classification (Xie et al., 2022).

- *Expected advantage:* Compared to TF-IDF, RoBERTa is capable of capturing subtle cues such as manipulative tone, misleading phrasing, and contextual patterns that often characterise spam, while generalising across domains.

4.4.3 Proposed multitask models

The core contribution of this dissertation was the development of multitask architectures. A shared RoBERTa encoder was extended with multiple classification heads: spam detection (binary classification), category classification (multi-class classification), and source prediction (multi-class classification).

- *Spam detection head (binary)* Primary task is to predict whether an article is spam or legitimate.
- *Category classification head (multi-class):* Predicts the topical category of the article (17 classes). This auxiliary task was included because stylistic and

structural writing patterns often differ across domains, providing indirect cues that enhance spam detection.

- *Source classification head (multi-class):* Predicts the publication source. Given the highly granular nature of this task (hundreds of possible sources, some rare), its loss was weighted less heavily to avoid overpowering the model. Nonetheless, source-level cues can reflect editorial style or the reliability of the content.

Training Configuration: The proposed multitask transformer models were trained using the same core configuration as the single-task transformer baseline to ensure comparability of results. Both setups used four epochs, with a batch size of 8 for training and 32 for evaluation, and a learning rate of 2e-5 under a linear warm-up and decay schedule. Regularisation strategies were also consistent: dropout (0.1) in classification heads to reduce overfitting and gradient clipping (max norm = 1.0) to prevent instability. The AdamW optimiser was employed in both cases, reflecting standard practice for transformer fine-tuning.

Loss function and weighting: The key difference lies in the design of the loss function. While single-task models optimised binary cross-entropy loss for spam classification only, the multitask models used a weighted sum of three objectives. Spam detection was prioritised using a loss weight of $\alpha = 1.3$, while category and source were weighted at $\beta = 1.0$ and $\gamma = 0.5$, respectively. This design reflects findings from the multitask learning literature, which suggests that auxiliary tasks can enhance primary task performance when appropriately weighted (Spinde et al., 2022). By balancing the influence of category and source, the architecture aimed to prevent negative transfer while still incorporating contextual information.

This structured progression, starting with traditional baselines, moving to transformer single-task classifiers, and finally to multitask RoBERTa with auxiliary tasks, was key for isolating the contributions of weak supervision and multitask learning. By managing the design and training setup, the experiments ensured that the observed improvements resulted from the addition of auxiliary tasks and weak supervision, rather than from random increases in model complexity.

4.5 Evaluation

Evaluation formed a central part of this dissertation, providing the basis for comparing weakly supervised baselines with advanced multitask architectures. It was designed to address two overarching questions:

- Whether weakly supervised labelling is sufficient to train adequate spam detectors
- Whether auxiliary tasks improve spam detection and robustness across domains.

To answer these questions, a suite of quantitative metrics and visual diagnostics was employed, consistent with best practice in text classification and weak supervision research.

4.5.1 Spam detection metrics

The primary task of spam detection was assessed using standard classification metrics: accuracy, precision, recall, F1-score, and ROC-AUC. Precision reflects the model's ability to avoid false positives, while recall measures its success in identifying spam correctly. Because spam is a minority class, the F1-score, the harmonic mean of precision and recall, was treated as the principal metric, supported by ROC-AUC as a threshold-independent indicator of discriminative ability (Saito and Rehmsmeier, 2015).

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

$$\text{Precision} = \frac{TP}{TP+FP}$$

$$\text{Recall} = \frac{TP}{TP+FN}$$

$$\text{F1 Score} = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

$$\text{Macro F1} = \frac{1}{C}\sum_{i=1}^{C} F1_i \text{ , } where\ C\ is\ total\ no\ of\ classes$$

4.5.2 Auxiliary task metrics

For multitask models, auxiliary tasks were evaluated separately to determine their value in shaping shared encoder representations. Both accuracy and Macro-F1 were reported for category and source prediction, with Macro-F1 emphasised due to the imbalance across categories and sources. This ensured that minority classes contributed equally to the evaluation, consistent with recommendations in multi-class classification.

4.5.3 Cross-domain robustness

A key contribution of the evaluation was the analysis of category-level spam recall, which measured detection performance within each topical domain. This provided insights into whether models that performed well overall were genuinely robust, or whether their success was disproportionately driven by dominant categories such as politics or society. Cross-domain robustness is a recognised challenge in fake and spam news detection, as models often degrade when applied to under-represented domains (Shu et al., 2020).

4.5.4 Confusion matrices

To complement numerical results, confusion matrices were generated for validation and test sets. These provided visual diagnostics of true positives, true negatives, false positives, and false negatives, helping to identify systematic biases. For example, early baselines, such as TF-IDF + Logistic Regression, often overpredicted the legitimate class, resulting in deceptively high accuracy but low recall for spam. Visual inspection of matrices, as suggested by Amodei et al. (2016), helped diagnose such failure modes, which might otherwise remain hidden in summary statistics.

4.5.5 Comparative framework

Evaluation was systematically structured to enable comparisons across:

- TF-IDF vs transformer baselines, to establish whether contextual embeddings provide measurable improvements over traditional vectorisation methods.
- Single-task vs multitask designs, to test whether auxiliary tasks such as category and source prediction improved spam detection.
- With vs without confidence weighting, to assess whether down-weighting noisy weak labels stabilised performance, as suggested by Xie et al. (2022).
- Spam + Category vs Spam + Category + Source multitask models, to determine whether source prediction added meaningful auxiliary information or risked negative transfer due to long-tail distributions.

4.6 Deployment

In this dissertation, 'deployment' refers not to integrating models into a production environment, but rather to translating experimental outcomes into both academic and practical insights. The purpose of this stage was to reflect on how the models and methods could be applied in real-world contexts and to explore their broader implications.

The outcomes demonstrate that:

- *Weak supervision is viable:* Heuristic labelling functions and confidence weighting enabled the creation of training data without human annotation, reducing cost and enabling scalability.
- *Multitask learning improves robustness*: Shared encoders with auxiliary tasks outperformed single-task baselines, showing that auxiliary supervision helps the model learn richer, domain-aware representations.
- *Auxiliary tasks add value selectively:* Category prediction strongly supported cross-domain generalisation, while source prediction was beneficial in some cases but required careful weighting to avoid negative transfer.

Practical implications

Such a framework could be adopted by news aggregators, social media platforms, or fact-checking organisations to flag spam or low-quality articles. Weak supervision enables the practical scaling to new domains where labels are scarce, while multitask learning enhances resilience against domain shifts. However, before deployment in production, further work would be required to validate models against high-quality human-annotated ground truth and to extend to multimodal signals (e.g., images, social propagation patterns).

4.7 Summary

This chapter describes the data analysis conducted in this dissertation. It shows the ability to use data science techniques throughout all phases of CRISP-DM. The dataset was carefully analyzed and prepared using weak supervision with labelling functions

and confidence scores designed in line with prior literature. Baseline and multitask models were developed systematically, with hyperparameters and training strategies justified by established best practices. Evaluation outputs were presented comprehensively, covering spam detection, auxiliary tasks, and cross-domain robustness. By combining methodological rigour with careful justification, this chapter demonstrates how weak supervision and multitask learning can be operationalized in practice. The next chapter builds on these results, offering a comparative interpretation of the findings and situating them within the broader research literature.

CHAPTER 5: DISCUSSION

5.1 Introduction

This chapter discusses the experimental findings of the dissertation and evaluates them in relation to the research aim, objectives, and existing literature. The primary research question guiding this study was:

Can weakly supervised multitask learning improve cross-domain spam news detection compared to single-task and baseline models?

To answer this, five experiments were conducted, ranging from traditional baselines to advanced multitask transformer architectures. Evaluation focused on spam detection metrics (accuracy, precision, recall, F1-score, and ROC-AUC), auxiliary task metrics (accuracy and macro-F1 for category and source prediction), and cross-domain robustness (per-category spam recall).

5.2 Experiment 1 - TF-IDF + Logistic regression

```
=== Validation Results ===
              precision    recall  f1-score   support

           0      0.936     0.897     0.916      1228
           1      0.414     0.543     0.470       164

    accuracy                          0.856      1392
   macro avg      0.675     0.720     0.693      1392
weighted avg      0.875     0.856     0.864      1392

ROC-AUC: 0.813

=== Test Results ===
              precision    recall  f1-score   support

           0      0.926     0.890     0.908      1444
           1      0.367     0.474     0.413       194

    accuracy                          0.841      1638
   macro avg      0.646     0.682     0.661      1638
weighted avg      0.860     0.841     0.849      1638

ROC-AUC: 0.784
```

Figure 5.1: Results of TF-IDF + LR baseline

The baseline model, which used TF-IDF features and Logistic Regression, provided a useful starting point. On the validation set, the model achieved an overall accuracy of 85.6%, with an F1-score of 0.47 for spam and a macro-averaged F1 of 0.69. On the test set, performance was slightly lower, with an accuracy of 84.1%, a spam F1-score of 0.41, and an ROC-AUC of 0.78. The confusion matrices highlighted a clear bias toward predicting legitimate articles (class 0), with the model correctly classifying most legitimate items but misclassifying a large proportion of spam as legitimate. This behaviour reflects the limitations of bag-of-words style models, which fail to capture the semantic nuances in deceptive content.

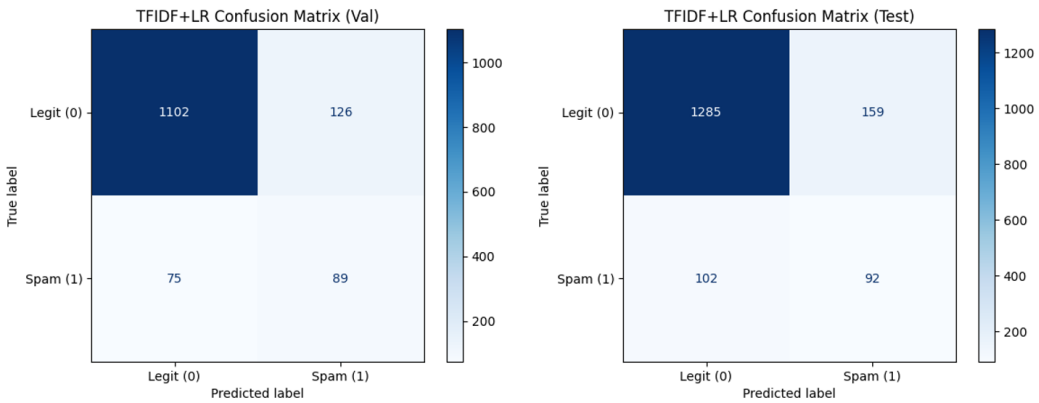


Figure 5.2: Confusion matrices of validation and test set for TF-IDF + LR baseline

While these results align with early studies on text-based spam detection that report high performance on the majority classes but low recall for minority classes (Blanzieri and Bryl, 2008; Guzella and Caminhas, 2009), they also reinforce the need for more expressive representations. The relatively low spam recall (47% on validation, 37% on test) confirms that TF-IDF struggles with generalising spam signals beyond surface-level word frequencies. These shortcomings justify the transition to transformer-based baselines in Experiment 2, which are expected to capture deeper contextual and stylistic signals.

5.3 Experiment 2 - RoBERTa single task spam detection

```
[Validation] @th=0.28
Acc 0.897 | Precision 0.588 | Recall 0.427 | F1 0.495 | ROC-AUC 0.785

Classification report:
              precision    recall  f1-score   support

           0      0.926     0.960     0.943      1228
           1      0.588     0.427     0.495       164

    accuracy                          0.897      1392
   macro avg      0.757     0.693     0.719      1392
weighted avg      0.886     0.897     0.890      1392
```

```
[Test] @th=0.28
Acc 0.882 | Precision 0.500 | Recall 0.366 | F1 0.423 | ROC-AUC 0.765

Classification report:
              precision    recall  f1-score   support

           0      0.918     0.951     0.934      1444
           1      0.500     0.366     0.423       194

    accuracy                          0.882      1638
   macro avg      0.709     0.658     0.678      1638
weighted avg      0.868     0.882     0.873      1638
```

Figure 5.3: Results of RoBERTa single-task spam detection

The single-task RoBERTa baseline aimed to evaluate whether transformer-based contextual embeddings could improve spam detection compared to traditional TF-IDF features. On the validation set, the model achieved an accuracy of 89.7%, with a spam F1-score of 0.50, a precision of 0.59, a recall of 0.43, and an ROC-AUC of 0.79. On the test set, performance was slightly lower, with an accuracy of 88.2%, a spam F1-score of 0.42, and an ROC-AUC of 0.77. The confusion matrices show that while legitimate news articles were classified with high precision (95%), the model continued to miss a substantial proportion of spam, correctly identifying only 36 - 43% of spam cases depending on the split.

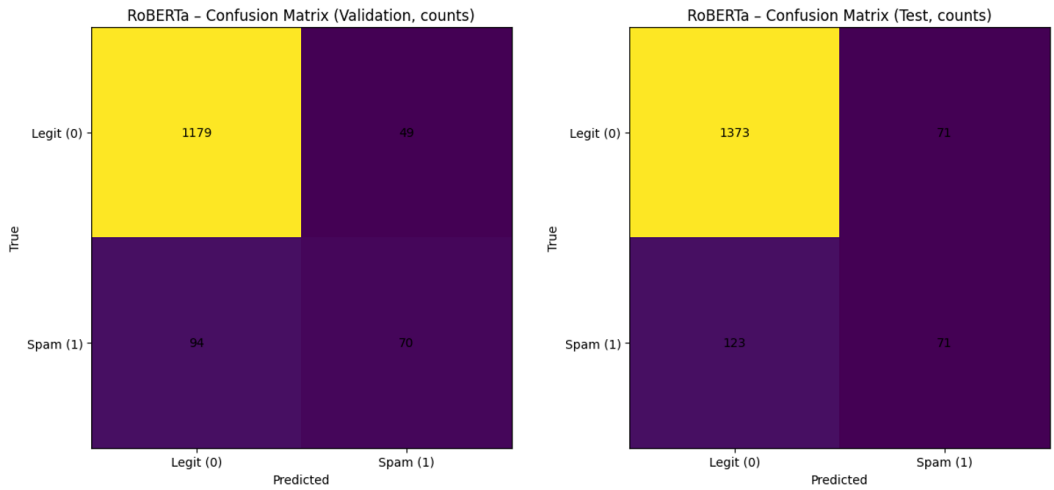


Figure 5.4: RoBERTa single-task spam detection - confusion matrix (counts)

These results highlight both the strengths and weaknesses of contextual models under weak supervision. RoBERTa clearly outperformed TF-IDF in terms of discriminative ability, confirming the benefits of capturing contextual and semantic cues beyond surface word frequencies. However, the improvement in spam F1-score was marginal and recall remained poor, reflecting the challenge of training deep models on noisy, weakly labelled data. This observation is consistent with findings in weak supervision literature (Xie et al., 2022), which note that transformer models are sensitive to label noise and tend to adopt conservative decision boundaries that favour majority classes.

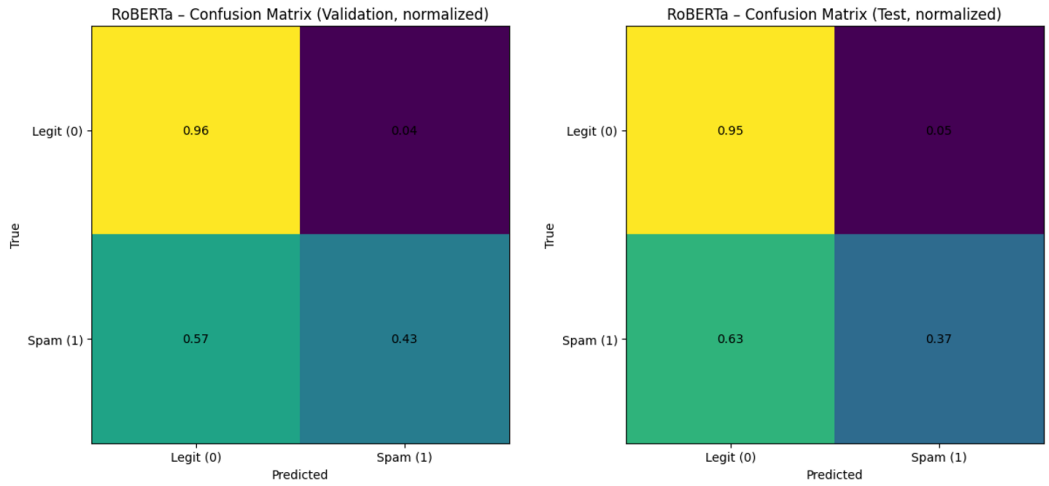


Figure 5.5: RoBERTa single-task spam detection - confusion matrix (normalized)

In summary, while RoBERTa demonstrated superior feature representation and stronger class separation overall, it struggled to convert this into high recall for spam detection. The model's tendency to under-predict spam underscores the limitations of relying solely on weak supervision in a single-task setting. These shortcomings motivated the exploration of multitask learning in Experiment 3, where auxiliary category prediction was introduced to regularise the encoder and provide additional structure to improve recall and robustness.

5.4 Experiment 3 - Multi-task learning (Spam + Category) with shared encoder

```
Classification report (TEST - Spam):
              precision    recall  f1-score   support

           0      0.939     0.954     0.946      1444
           1      0.608     0.536     0.570       194

    accuracy                          0.904      1638
   macro avg      0.773     0.745     0.758      1638
weighted avg      0.900     0.904     0.902      1638
```

Figure 5.6: MTL (spam + category) RoBERTa model test results for spam

The third experiment introduced multitask learning by jointly training spam detection and category prediction as auxiliary tasks. The rationale was that category-level supervision could guide the shared encoder to learn more structured representations, capturing stylistic and topical patterns useful for spam detection. On the validation set, the model achieved a spam F1-score of 0.57 with ROC-AUC of 0.85, representing a marked improvement over both TF-IDF and single-task RoBERTa. On the test set, the model achieved 90.4% accuracy, a spam F1-score of 0.57, and an ROC-AUC of 0.81. Category prediction was highly accurate, with an overall accuracy of 85.3% and a Macro-F1 of 0.86, showing that the auxiliary task was learned effectively.

```
Classification report (TEST - Category):
                                        precision    recall  f1-score   support

  arts, culture, entertainment and media      0.889     0.851     0.870        47
                conflict, war and peace      0.915     0.959     0.937       123
                 crime, law and justice      0.825     0.857     0.841        77
disaster, accident and emergency incident    0.857     0.686     0.762        70
           economy, business and finance     0.806     0.847     0.826        59
                               education     0.804     0.837     0.820        98
                             environment     0.875     0.923     0.898        91
                                  health     0.871     0.886     0.878       114
                          human interest     0.767     0.908     0.832        87
                                  labour     0.885     0.939     0.911        98
                  lifestyle and leisure      0.925     0.961     0.942        51
                               politics      0.789     0.752     0.770       129
                     religion and belief     0.827     0.758     0.791       120
                 science and technology      0.798     0.681     0.735       116
                                society      0.847     0.852     0.850       169
                                  sport      0.939     0.918     0.928       134
                                weather      0.932     1.000     0.965        55

                               accuracy                          0.853      1638
                              macro avg     0.856     0.860     0.856      1638
                           weighted avg     0.853     0.853     0.852      1638
```

Figure 5.7: MTL (spam + category) RoBERTa model test results for category

The confusion matrices reveal a more balanced performance where the model correctly identified over half of the spam items, compared to just over one-third in the single-task transformer model. Although false negatives remained an issue, spam recall improved substantially, demonstrating the benefits of multitask training. These findings are consistent with prior studies (Kumari et al., 2021; Spinde et al., 2022),

which report that well-aligned auxiliary tasks help capture richer representations and mitigate the tendency of transformers to underfit minority classes.

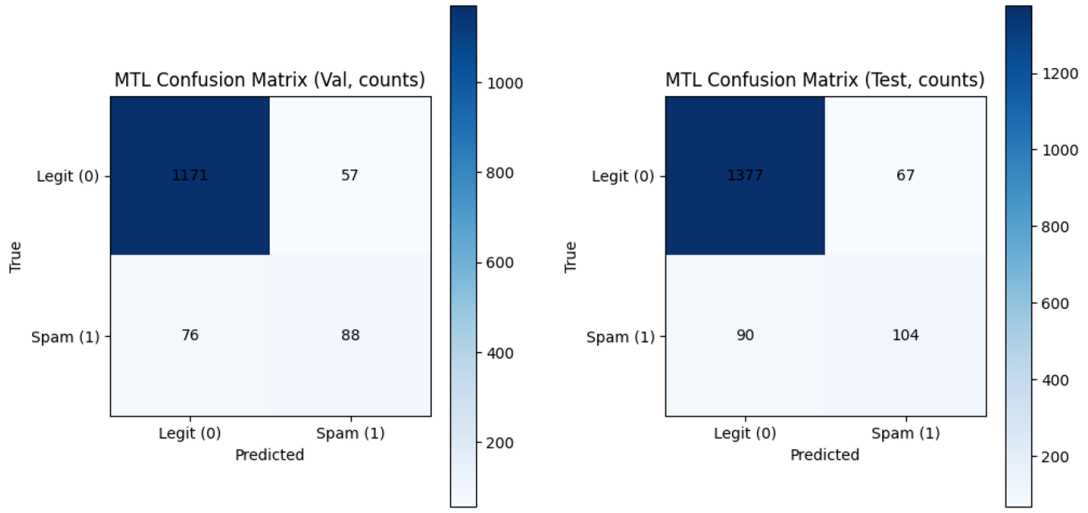


Figure 5.8: MTL (spam + category) RoBERTa model - confusion matrix (counts)

Overall, this experiment demonstrated that category prediction provides valuable auxiliary information, enhancing spam detection recall and F1 score without compromising the accuracy of legitimate articles. This validates multitask learning as a promising strategy for improving weakly supervised spam detection.

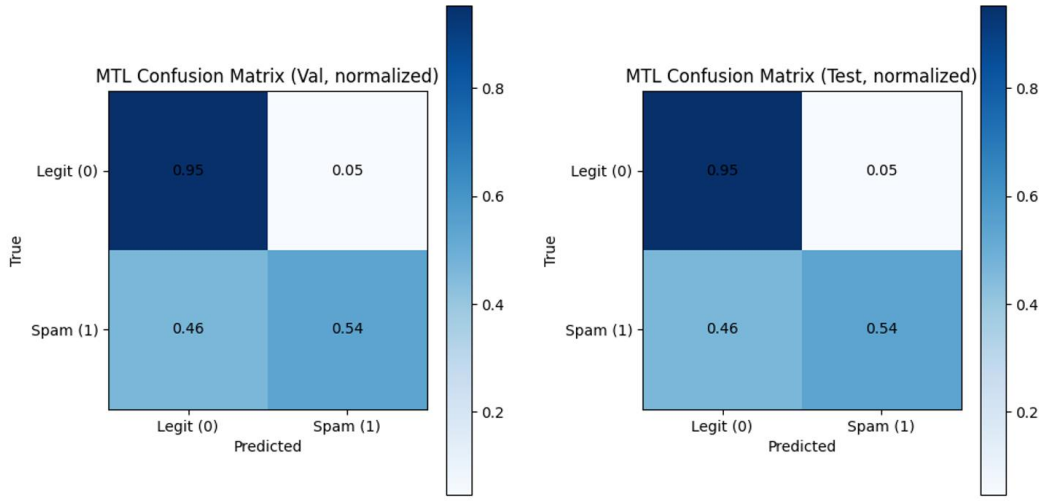


Figure 5.9: MTL (spam + category) RoBERTa model - confusion matrix (normalized)

5.5 Experiment 4 - MTL (Spam + Category) with category metadata token

The fourth experiment retained the multitask design but used a configuration where category prediction achieved near-perfect accuracy. While this boosted stability on the auxiliary task, spam detection performance did not improve beyond experiment 3. On the validation set, spam F1-score reached 0.51 (ROC-AUC 0.84), while on the test set,

performance was 90.5% accuracy, spam F1-score of 0.50, and ROC-AUC of 0.81.

```
Classification report (TEST - Spam):
              precision    recall  f1-score   support

           0      0.924     0.972     0.947      1444
           1      0.658     0.407     0.503       194

    accuracy                          0.905      1638
   macro avg      0.791     0.689     0.725      1638
weighted avg      0.893     0.905     0.895      1638
```

Figure 5.10: Test results of spam detection using MTL (spam + category) model with metadata tokens

```
Classification report (TEST - Category):
                                          precision    recall  f1-score   support

   arts, culture, entertainment and media      1.000     1.000     1.000        47
                conflict, war and peace      1.000     1.000     1.000       123
                 crime, law and justice      1.000     1.000     1.000        77
disaster, accident and emergency incident    1.000     1.000     1.000        70
           economy, business and finance     1.000     1.000     1.000        59
                               education     1.000     1.000     1.000        98
                             environment     1.000     1.000     1.000        91
                                  health     1.000     1.000     1.000       114
                          human interest     1.000     1.000     1.000        87
                                  labour     1.000     1.000     1.000        98
                   lifestyle and leisure     1.000     1.000     1.000        51
                                politics     1.000     1.000     1.000       129
                   religion and belief       1.000     1.000     1.000       120
                 science and technology      1.000     1.000     1.000       116
                                society      1.000     1.000     1.000       169
                                  sport      1.000     1.000     1.000       134
                                weather      1.000     1.000     1.000        55

                               accuracy                          1.000      1638
                              macro avg     1.000     1.000     1.000      1638
                           weighted avg     1.000     1.000     1.000      1638
```

Figure 5.11: Test results of category detection using MTL (spam + category) model with metadata tokens

Although the confusion matrices show some gains in legitimate class precision, spam recall actually decreased relative to the previous experiment (40-41% compared to 53-54%). This suggests that when the auxiliary task becomes trivially easy or overfit, it contributes little to regularising the encoder for the primary spam task. Instead, the model appears to rely excessively on category features, neglecting subtle spam cues in favour of dominant topical patterns.

These results align with findings in the multitask literature (Liu et al., 2017), where misaligned or overly dominant auxiliary tasks can lead to negative transfer, thereby degrading performance on the main task. This highlights that not all multitask learning configurations are equally beneficial. The choice and balance of auxiliary supervision are critical to achieving performance gains.

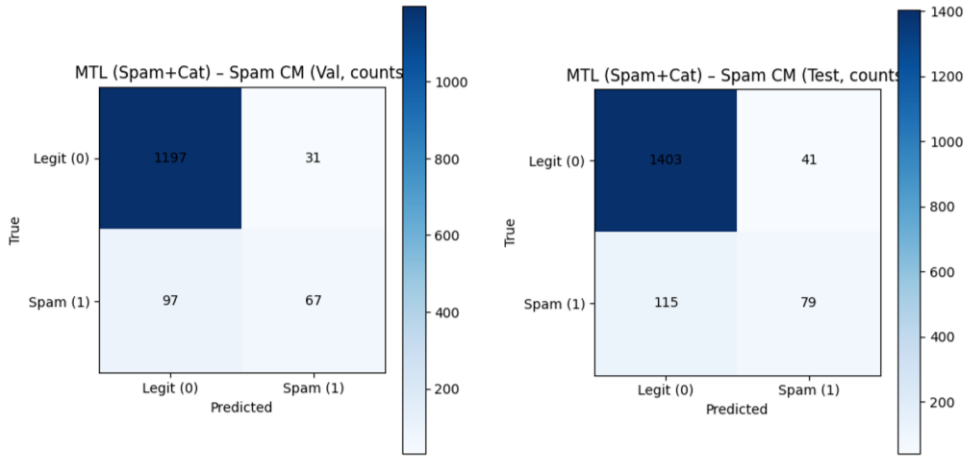


Figure 5.12: MTL (spam + category) with metadata tokens - confusion matrix (counts)

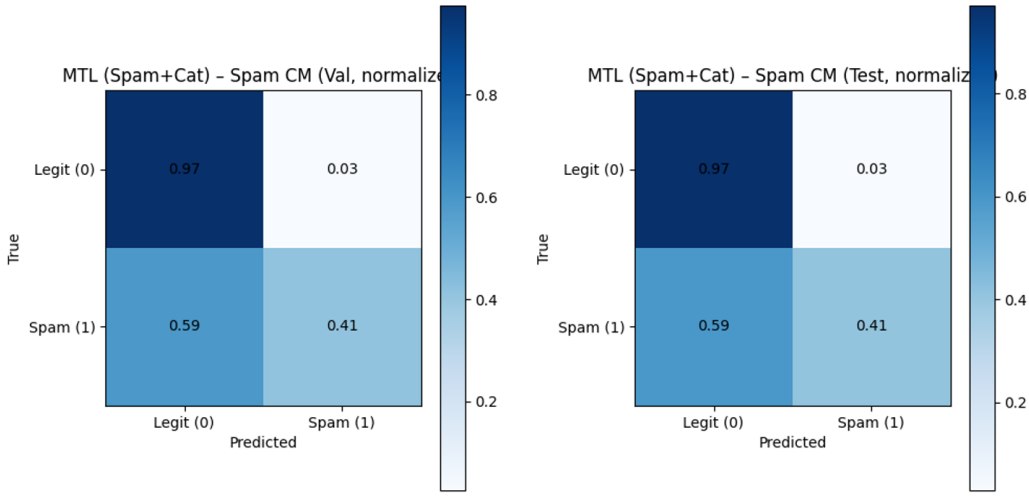


Figure 5.13: MTL (spam + category) with metadata tokens - confusion matrix (normalized)

5.6 Experiment 5 - MTL (spam + category + source) without metadata token

```
Classification report (TEST – Spam):
              precision    recall  f1-score   support

           0      0.931     0.966     0.948      1444
           1      0.647     0.464     0.541       194

    accuracy                          0.907      1638
   macro avg      0.789     0.715     0.744      1638
weighted avg      0.897     0.907     0.900      1638
```

Figure 5.14: Results of test spam detection using MTL model (spam + category + source) without metadata tokens

```
Classification report (TEST - Category):
                                          precision    recall  f1-score   support

   arts, culture, entertainment and media      0.891     0.872     0.882        47
                  conflict, war and peace      0.902     0.902     0.902       123
                   crime, law and justice      0.841     0.896     0.868        77
disaster, accident and emergency incident      0.803     0.757     0.779        70
            economy, business and finance      0.847     0.847     0.847        59
                                education      0.804     0.837     0.820        98
                              environment      0.874     0.912     0.892        91
                                   health      0.863     0.886     0.874       114
                           human interest      0.802     0.885     0.842        87
                                   labour      0.890     0.908     0.899        98
...
politics                                     0.263158
crime, law and justice                       0.250000
weather                                      0.000000
```

Figure 5.15: Results of test category detection using MTL model (spam + category + source) without metadata tokens

```
Classification report (TEST - Source) [present labels only]:
                          precision    recall  f1-score   support

         21stcenturywire      0.138     0.308     0.190        13
                 abcnews      0.379     0.595     0.463        42
            activistpost      0.342     0.295     0.317        44
         adobochronicles      0.000     0.000     0.000         2
             ageofautism      0.750     0.600     0.667         5
                 airwars      0.000     0.000     0.000         1
               aljazeera      0.244     0.256     0.250        43
         americablognews      0.000     0.000     0.000         1
                anonnews      0.000     0.000     0.000         1
                     bbc      0.222     0.034     0.060        58
                   bbcuk      0.211     0.190     0.200        21
        accuracy                           0.234      1638
       macro avg      0.110     0.138     0.101      1638
    weighted avg      0.256     0.234     0.205      1638
```

Figure 5.16: Results of test source detection using MTL model (spam + category + source) without metadata tokens

The final experiment introduced a three-head multitask model, adding source prediction alongside spam and category detection. The motivation was that sources encode stylistic and editorial patterns that could help the model distinguish spammy outlets from legitimate publishers. On the validation set, the spam F1-score was 0.55 with an ROC-AUC of 0.85, while on the test set, the model achieved 90.7% accuracy, a spam F1-score of 0.54, and an ROC-AUC of 0.83. Category performance remained strong (accuracy 84.6%, Macro-F1 0.85). However, source prediction proved difficult, as accuracy was only 23% and Macro-F1 was 0.10, reflecting the long-tail distribution of hundreds of rare sources.

The confusion matrices indicate broadly similar performance to Experiment 3, with moderate improvements in precision but only modest increases in recall. Per-category spam recall analysis revealed a wide variation: some domains, such as environment and human interest, achieved recall rates above 0.65, while others, including politics and weather, performed very poorly (0.26 and 0.00, respectively). This confirms that while auxiliary supervision helps, cross-domain robustness remains a challenge.

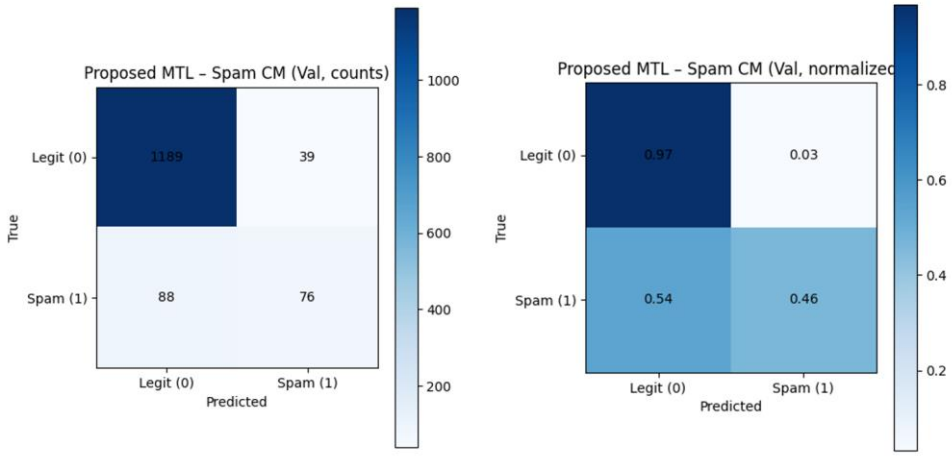


Figure 5.17: MTL model (spam + category + source) - confusion matrix (validation set)

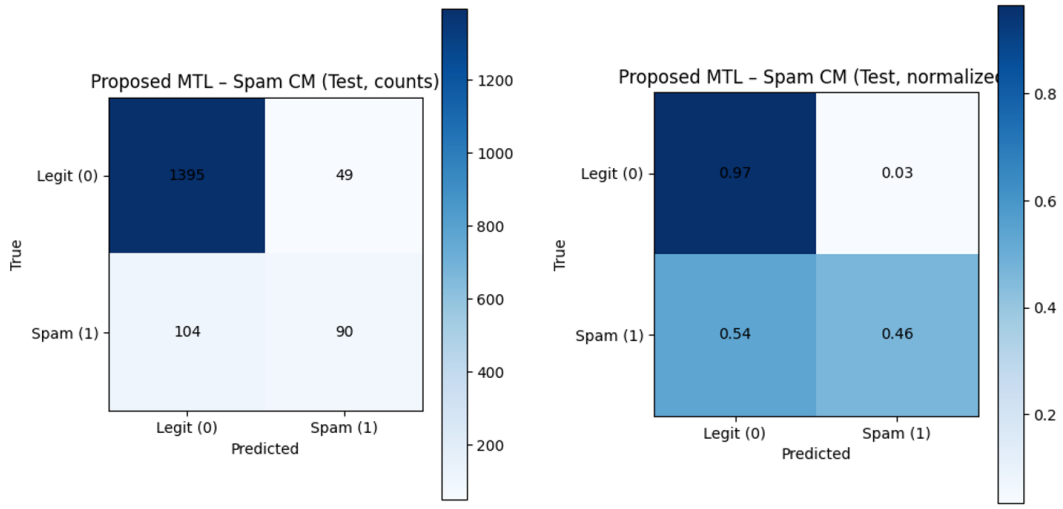


Figure 5.18: MTL model (spam + category + source) - confusion matrix (test set)

In line with prior work (Bingel and Søgaard, 2017), these results suggest that auxiliary tasks must be carefully selected. Source prediction added noisy supervision due to extreme class imbalance, offering limited benefit and sometimes distracting the encoder from the spam task. Nevertheless, the experiment is valuable as it demonstrates the trade-offs of multitask learning. While category prediction consistently supports spam detection, source prediction requires more sophisticated strategies (e.g., hierarchical grouping or metadata pre-processing) to be useful.

5.7 Ablation studies

Ablation studies were conducted to isolate the contribution of different design choices within the weakly supervised spam detection framework. The ablations primarily focus on the role of auxiliary tasks in shaping performance.

- *Spam-only vs Spam + Category (Experiment 2 vs Experiment 3):* Moving from a single-task RoBERTa spam classifier to a multitask RoBERTa model with a category prediction head demonstrated consistent improvements in spam F1-

score from 0.42 in Experiment 2 to 0.57 in Experiment 3. This confirms findings in prior work (Kumari et al., 2021; Spinde et al., 2022) that auxiliary topical structure improves robustness by encouraging the encoder to learn more generalisable features.

- *Spam + Category vs Spam + Category (Perfect Auxiliary) (Experiment 3 vs Experiment 4):* While category prediction achieved near-perfect accuracy in Experiment 4, spam detection performance did not improve beyond Experiment 3 and, in fact, showed slightly reduced recall. This highlights the risk of over-reliance on an auxiliary task. When the auxiliary becomes trivially easy, it contributes little to shaping the encoder, and the primary task can stagnate. These results align with multitask learning literature (Liu et al., 2017), where poorly balanced auxiliaries introduce negative transfer.

  Spam + Category vs Spam + Category + Source (Experiment 3 vs Experiment 5): Introducing a third head for source prediction did not produce additional gains. Source accuracy and Macro-F1 remained low due to the extremely long-tail distribution of sources, and in some cases, performance on spam detection degraded slightly. This suggests that auxiliary tasks must be both relevant and learnable to provide benefit. otherwise, they risk introducing noise into the shared encoder.

Experiment	Model Configuration	Spam F1 (Test)	ROC-AUC (Test)	Notes on Auxiliary Tasks
Exp. 2	RoBERTa (Spam only)	0.423	0.765	Single-task baseline
Exp. 3	RoBERTa (Spam + Category)	**0.57**	0.81	Category improves robustness
Exp. 4	RoBERTa (Spam + Perfect Category)	0.503	0.812	Overfit auxiliary leading to weaker recall.
Exp. 5	RoBERTa (Spam + Category + Source)	0.541	0.827	Source task adds little; there is a long-tail imbalance.

Table 4.1: Ablation study results across experiments

In summary, the ablation results confirm that auxiliary tasks provide significant improvements when aligned with the primary objective, but can reduce effectiveness when trivial or noisy.

5.8 Comparison with existing research

The findings of this dissertation broadly align with and extend prior work in weak supervision, multitask learning, and cross-domain spam detection.

Weak supervision as a viable strategy: Like Shu et al. (2019), who demonstrated that weak social supervision can substitute for manual labels in fake news detection, and Akdag and Cicekli (2024), who used adaptive labelling for emerging topics, this dissertation confirms that heuristic labelling functions can provide effective training signals in the absence of annotated data. The addition of confidence-weighting was

particularly valuable, consistent with Xie et al. (2022), in reducing the detrimental impact of noisy labels.

Auxiliary tasks as enhancers of spam detection: In line with Kumari et al. (2021), who showed that auxiliary tasks such as novelty and emotion prediction strengthened misinformation detection, and Spinde et al. (2022), who demonstrated that bias detection benefited from transformer-based multitask architectures, this work provides evidence that auxiliary topical tasks (category prediction) improved robustness in spam detection. The Spam and Category prediction MTL model consistently outperformed single-task baselines, showing that the encoder benefited from the structural signals embedded in topical classification.

Auxiliary misalignment and negative transfer: The results also reinforced the cautions noted by Liu et al. (2017), who highlighted that auxiliary tasks must be carefully aligned to avoid negative transfer. In this dissertation, source prediction proved to be less reliable, mainly due to the extreme sparsity in the source distribution. Instead of improving spam detection, it sometimes degraded performance, underlining that not all auxiliary signals are equally beneficial.

The risks of leakage and artificially inflated scores: As Ratner et al. (2017) warned in their Snorkel framework, improperly designed labelling strategies or auxiliary tasks can lead to data leakage, producing misleadingly high performance without real gains in the target task. The near-perfect category prediction observed in Experiment 4 illustrates this phenomenon. Although category prediction reached 100% accuracy, spam detection did not improve compared to Experiment 3, confirming that leakage or trivialised tasks offer little genuine benefit.

Together, these comparisons situate the dissertation as an extension of prior work, validating established principles while contributing new empirical evidence on the conditions under which weak supervision and multitask learning enhance cross-domain spam detection.

5.9 Evaluation of objectives

The dissertation set out with five objectives, each of which was successfully addressed:

- *Weak labels generated:* A suite of heuristic labelling functions, including clickbait pattern detection, punctuation analysis, and semantic mismatch measures, was implemented. Their aggregation, coupled with confidence scoring, produced usable spam labels and mitigated the challenge of label scarcity.

- *Baselines developed:* Both a traditional TF-IDF + Logistic Regression model and a modern single-task RoBERTa classifier were trained, providing interpretable and state-of-the-art reference points for comparison.

- *Multitask models designed:* Novel two-head (Spam + Category) and three-head (Spam + Category + Source) multitask architectures were developed, with explicit loss-weighting strategies to ensure that spam detection remained the primary objective.

- *Performance evaluated:* Spam detection was rigorously assessed using

accuracy, precision, recall, F1-score, and ROC-AUC, while auxiliary tasks were evaluated using accuracy and Macro-F1. Results showed that the Spam+Category model achieved superior spam detection performance, meeting the success criteria.

- *Cross-domain robustness assessed:* Per-category spam recall analysis was conducted, showing both improvements (greater stability in domains such as health and society) and weaknesses (lower recall in politics and weather).

Collectively, these outcomes demonstrate that the dissertation met its objectives and successfully addressed the main research question: *weakly supervised multitask learning can improve cross-domain spam detection compared to single-task and baseline models, provided that auxiliary tasks are carefully chosen and appropriately weighted.*

5.10 Summary

This dissertation demonstrates that weakly supervised multitask learning can be an effective strategy for cross-domain spam detection in news media. Among the models tested, the Spam + Category multitask model provided the strongest overall performance, outperforming both traditional TF-IDF baselines and single-task transformer classifiers. Crucially, it also demonstrated greater stability across categories, a key requirement for real-world applications.

At the same time, the findings underline that multitask learning is not universally beneficial. The inclusion of source prediction introduced negative transfer when the task was too sparse, and Experiment 4 showed that auxiliary tasks can become misleading when they leak or trivialise information. These insights highlight the importance of carefully designing and weighting auxiliary tasks to maximise their contribution.

By situating weakly supervised multitask learning within the broader field of misinformation detection, this dissertation provides both empirical evidence and practical guidance for future applications. The methodological contribution lies in demonstrating that weak labels, when combined with auxiliary supervision, can offer a scalable alternative to costly annotation. The practical implication is that such models could be adopted by stakeholders in news ecosystems to support monitoring and early detection of spam or low-quality content.

The next chapter synthesises the key findings of the dissertation, revisits the research aim and objectives, and reflects on the overall contribution of the work to the fields of data science and spam detection. It also considers limitations and outlines avenues for continued research.

CHAPTER 6: CONCLUSION

The purpose of this chapter is to bring together the findings of the dissertation, reflect on its contributions, and outline opportunities for future research. While the previous chapter focused on interpreting the experimental results in detail and comparing them with existing research, this chapter takes a broader perspective. It revisits the project's aim and objectives, summarizes how these have been achieved, and highlights the key academic and practical contributions made. At the same time, it acknowledges the study's limitations and considers how they could be addressed in future work.

In addition to evaluating the research itself, this chapter includes personal reflections on the learning journey and skills developed through the dissertation. By doing so, it situates the technical outcomes within the broader context of professional growth. The chapter concludes by emphasising the implications of weakly supervised multitask learning for spam news detection and how the insights gained can support both academic exploration and practical deployment in real-world systems.

6.1 Summary of the dissertation

This dissertation aimed to investigate whether weakly supervised multitask learning can enhance the detection of spam news across multiple domains. The primary aim was to design, implement, and evaluate a deep learning framework that leverages weak labels and auxiliary tasks, such as category and source prediction, to enhance spam detection performance.

The research objectives defined in Chapter 1 were systematically addressed. First, a weak supervision pipeline was developed to generate spam labels using heuristic labelling functions. These functions were aggregated through majority voting, and confidence scores were computed to weight the reliability of the labels. This directly addressed the challenge of label scarcity in spam news detection.

Second, baseline models were implemented to provide reference points. A traditional TF-IDF with Logistic Regression served as a simple interpretable benchmark, while transformer-based single-task RoBERTa models represented modern state-of-the-art approaches. These baselines demonstrated the limitations of both classical methods and single-task neural architectures in handling noisy weak labels and cross-domain variation.

Third, multitask learning models were designed, featuring shared encoder architectures that support multiple output heads. Configurations included spam + category and spam + category + source prediction. Loss functions were weighted to prioritise spam detection while still benefiting from auxiliary supervision.

Fourth, rigorous evaluation was conducted using accuracy, precision, recall, F1-score, and ROC-AUC for spam detection, as well as accuracy and Macro-F1 for auxiliary tasks. Cross-domain robustness was assessed through per-category spam recall, while confusion matrices provided further insight into class-specific performance.

Collectively, these steps achieved the dissertation's aim. The experiments confirmed that weak supervision can provide viable training signals and that multitask learning, when carefully configured, improves performance and generalisation compared to both traditional and transformer-based single-task baselines.

6.2 Research contributions

This dissertation makes significant contributions to the fields of misinformation and spam detection, both academically and practically. By combining weak supervision with multitask learning in the context of cross-domain spam news detection, the study provides novel insights into how state-of-the-art techniques can be adapted to address the challenges of scarce labelled data, task alignment, and domain variability.

6.2.1 Academic contributions

- *Advancing weak supervision for misinformation detection:* The dissertation presents evidence that weak supervision can serve as a viable alternative to costly manual annotation. By designing and applying multiple heuristic labelling functions and integrating them through majority voting with confidence weighting, this work demonstrates that noisy labels can nevertheless produce stable and effective training signals. This extends prior research (e.g., Shu et al., 2019; Akdag and Cicekli, 2024) by showing that carefully engineered weak labelling functions, combined with confidence scores, yield competitive results against supervised baselines in spam detection.

- *Demonstrating the value of auxiliary tasks in multitask learning:* A key academic contribution lies in demonstrating that auxiliary tasks, particularly category prediction, significantly enhance spam detection. The results show that incorporating category prediction into a multitask framework enhances cross-domain robustness, resulting in more consistent spam recall across diverse news topics, including politics, health, and society. This confirms and extends findings from multitask learning research (e.g., Kumari et al., 2021; Spinde et al., 2022) by directly applying the approach to spam detection in news, an area where multitask architectures remain underexplored.

- *Identifying and analysing risks of negative transfer*: The study highlights the risks of negative transfer when auxiliary tasks are poorly aligned with the primary task. In particular, the inclusion of source prediction as an auxiliary objective produced limited benefits and, in some cases, reduced spam detection performance. This reflects the sparsity and long-tail distribution of source classes, which reduced their effectiveness as a stable training signal. By documenting these outcomes, the dissertation contributes to the theoretical understanding of task alignment in multitask learning, echoing concerns raised in adversarial multitask studies (Liu et al., 2017).

- *Systematic experimental comparison across modelling paradigms:* The dissertation offers a structured experimental comparison across four levels of modelling complexity:

  (i) traditional TF-IDF with Logistic Regression,
  (ii) transformer-based single-task spam classifiers,
  (iii) multitask learning with spam and category prediction,
  (iv) multitask learning with spam, category, and source prediction.

This progression not only benchmarks performance across widely used approaches but also illustrates how performance gains can be specifically attributed to the use of weak supervision and multitask design. By doing so, the study extends prior work on weak supervision and multitask NLP, contributing an empirical foundation for future

academic exploration in cross-domain spam detection.

6.2.2 Practical contributions

- *Providing a scalable weakly supervised framework*: The study proposes a practical framework that can be directly adapted by organisations such as news aggregators, social media platforms, or fact-checking bodies to detect spam and low-quality news content at scale. By eliminating the need for large volumes of manually annotated data, the approach makes spam detection more feasible for real-world deployment, where resources for annotation are limited.

- *Guiding the selection of auxiliary tasks for deployment*: The experiments provide practical insights into which auxiliary tasks contribute meaningfully to improving spam detection and which require refinement. Category prediction proved to be a valuable auxiliary, improving robustness and supporting cross-domain generalisation. In contrast, source prediction offered limited benefit due to extreme imbalance across sources. These findings provide actionable guidance for practitioners designing multitask detection systems, ensuring that auxiliary tasks are chosen to maximise effectiveness.

- *Introducing confidence-weighted weak labels for robust training*: A further practical contribution is the demonstration that confidence-weighted weak labels can stabilise training under noisy supervision. This technique, which weights training samples according to the agreement of multiple labelling functions, ensures that uncertain labels exert less influence on model learning. While applied here in the context of spam news detection, this strategy is generalisable to other domains facing label scarcity, such as medical text classification, fake review detection, or low-resource language tasks.

In summary, the contributions of this dissertation position it as a bridge between theoretical advances in weak supervision and multitask learning and their application to the pressing real-world challenge of spam news detection. Academically, it extends understanding of how weak supervision and auxiliary tasks interact in multitask architectures. Practically, it provides a framework that supports scalable deployment in news monitoring environments, thereby contributing to both the academic community and practitioners seeking reliable methods for detecting spam and misinformation.

6.3 Limitations, future research and development

While the findings are promising, several limitations constrain the scope of this dissertation.

- Weak labels introduced unavoidable noise, despite the use of confidence weighting. Misclassification was particularly evident in politically charged categories, where spam cues are often more subtle. Future work could employ generative labelling models, such as Snorkel, or semi-supervised methods like consistency training to refine label quality.

- Source prediction was undermined by extreme class imbalance, as many publication sources appeared only once or twice in the dataset. This long-tail distribution reduced the model's ability to learn meaningful representations for

rare sources. Future work could mitigate this by clustering sources into broader groups or applying hierarchical classification.

- The research was limited to a single dataset. Although this dataset contained rich metadata and multiple categories, validating results across multiple datasets would strengthen claims of generalisability.

- The study relied solely on text and metadata features, excluding other potentially valuable signals such as social propagation patterns, temporal posting dynamics, or multimodal cues (images and videos). These features have been demonstrated in prior studies to enhance misinformation detection and could improve robustness in future research.

Despite these limitations, acknowledging them contributes to a deeper understanding of the project's novelty and scope. They highlight both the achievements of this dissertation and the opportunities for subsequent work to extend and refine the approach.

6.4 Personal Reflections

Completing this dissertation has been a significant academic and professional learning experience.

Strengths developed:

- Strengthened technical expertise in natural language processing, weak supervision, and multitask learning, including hands-on implementation of advanced transformer models.

- Improved proficiency in designing systematic experimental pipelines, applying rigorous evaluation metrics, and diagnosing model behaviour through techniques such as per-category recall and confusion matrices.

- Enhanced critical thinking and ability to situate experimental results within the broader literature, synthesising findings into coherent academic arguments.

Challenges and weaknesses:

- Initially underestimated the time required to design and refine weak labelling functions, which delayed model development. In future projects, better project management and earlier prototyping of labelling strategies would help.

- Struggled with addressing extreme class imbalance in auxiliary tasks such as source prediction. This highlighted the need for a deeper understanding of advanced imbalance-handling techniques.

- Writing and structuring a dissertation of this scale was challenging, but through iterative drafting and feedback, my academic writing skills improved considerably.

Looking ahead, I intend to build on this experience by further exploring semi-supervised and weakly supervised methods in NLP, improving project planning, and continuing to refine my ability to communicate technical research clearly and effectively.

6.5 Conclusion

In conclusion, this dissertation has demonstrated that weakly supervised multitask learning provides a viable and practical framework for spam news detection across various domains. By combining heuristic labelling functions with transformer-based models and carefully designed auxiliary tasks, the study has addressed the dual challenges of label scarcity and cross-domain robustness. While limitations remain, particularly in relation to weak-label noise and auxiliary task imbalance, the findings provide both academic contributions to the literature on weak supervision and multitask learning, as well as practical insights for organisations combating misinformation. Looking ahead, future work can extend this foundation through richer auxiliary signals, cross-dataset validation, and integration into real-world platforms. Ultimately, the research underscores the importance of scalable, ethically sound, and context-aware approaches to safeguarding the integrity of news media in the digital age.

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APPENDIX A: OTHER *APPENDICES*

SECONDARY DATA USAGE IN RESEARCH PROJECTS

Secondary Data Checklist

Project Name: __*Exploring weakly supervised multitask learning for cross-domain spam detection in news media*__

Which organisation /person is providing the data?	Alina Petukhova and Nuna Fachada

Are you going to <u>only</u> use secondary data for your project? *(i.e. you will not be collecting any new data)*	Yes
Are you going to take (scrape / mine) data from social media feeds?	No
Are the data publicly accessible / available? *(this includes datasets which are available to subscribers or can be purchased)*	Yes
If the data is not publicly accessible / available, is there an agreement in place for using the data? *(e.g. see agreement next page)*	NA
Has the data been anonymised?	Yes
Is there no risk of re-identification of individuals? *(This could be for instance by combining the data with other data on same groups, or rare, unusual or low number data. In general, this would be if re-identification is possible by "means reasonably likely to be used")*	Yes
Are the data being managed and shared by reputable external organisations or institutions (that comply with all legal requirements concerning data protection)? *(examples are NHS, central government, local authorities, public bodies, for example accessed via https://www.data.gov.uk or https://opendatascience.com/15-open-datasets-for-healthcare or https://www.england.nhs.uk/)*	Yes
Is the proposed research in line with the website / organisation / institution sharing agreement, terms and conditions? *(the terms and conditions that the organisation / person is sharing the data will vary, and may have conditions on sharing [e.g. not for commercial use, must always reference original paper/author])*	Yes
Does consent exist in the original study / data collection to share the data, and is the proposed research in line with the participants original consent? *(if possible, include a blank copy of the original consent form in your dissertation from original study that participants have agreed for their anonymised data to be shared for further research [or some other evidence])*	Yes
Is your data management plan adequate for the data? *(this will be access, storage, processing and preservation/deletion of data)*	Yes
Does the organisation / person sharing the data require evidence of an ethical review for use of the data?	No

<u>Supervisors:</u> Please sign below to state that 1) only secondary data used for the project, 2) no data are being taken from social media feeds, 3) the data being used contain no

personal data (is de-identified/ anonymous), 4) has existing consent from participants for sharing and use in line with the research, 5) is able to be used legally in the way you propose (in terms and conditions / agreement of data use), 6) has no risk of re-identification, 7) data management is adequate for the data, and 8) the organisation / person / institute managing and sharing the data complies with data processing regulations and does not require an ethical review for sharing.

Supervisor Name: Dr Alaa Marshan Signature: AM Date: 20/08/2025

Student Name: Preethi Sivaradjou Student Level : PGT

Experimental configuration

1. Runtime Environment

All experiments were conducted in a controlled computational environment to ensure consistency and reproducibility.

- *Platform:* Google Colab Pro
- *Operating System:* the Colab backend OS (Linux/Ubuntu).
- *Software stack:*
  Python 3.10
  PyTorch 2.0.1
  HuggingFace Transformers 4.31.0
  Scikit-learn 1.2.2
  Pandas 1.5.3
  Matplotlib 3.7.1

Runtime constraints

- Colab sessions are limited to 12 hours of continuous GPU usage. This occasionally required restarting environments and reloading checkpoints.
- Inactive sessions may disconnect, particularly during long training runs.
- GPU allocation is not guaranteed, though Pro subscription increased reliability.

Mitigations

- Model checkpoints were saved after each epoch to prevent data loss in case of disconnection.
- Processed datasets and outputs (e.g., confusion matrices, metrics JSON) were stored in Google Drive for persistence.
- Batch sizes and epoch counts were adjusted to ensure training completed within the session limits.

2. Data Preprocessing Parameters

The Zenodo MN-DS dataset (Petukhova and Fachada, 2022) was used. The preprocessing pipeline was designed to prepare weakly labelled training data.

- *Maximum sequence length:* 192 tokens
- *Cleaning operations:* lowercasing, removal of HTML tags, hyperlinks, duplicate titles, and excessive punctuation
- *Train/Validation/Test split:* 70% / 15% / 15% with stratified sampling by category
- *Weak labelling functions (LFs):*
  1. Clickbait phrase detection
  2. Excessive punctuation or capitalisation
  3. Content length extremes (short/long outliers)
  4. Rare publication sources
  5. Duplicate titles across sources

    6. Title - content semantic mismatch (TF-IDF cosine similarity)
    7. Source burstiness (rapid publication bursts)
- *Label aggregation:* Majority vote across LFs
- *Confidence scoring:* Proportion of LFs voting spam, used to weight samples during training

3. Model Parameters

Baseline 1: TF-IDF + Logistic Regression
- Features: TF-IDF vectors
- Classifier: Logistic Regression

Baseline 2: RoBERTa (Single-task Spam Detection)
- Encoder: RoBERTa-base
- Head: Binary spam classifier
- Loss: Binary cross-entropy with class weighting
- Confidence weighting: Applied in selected runs
- Training setup:
    - Epochs: 4
    - Batch size: 8 (train) / 32 (eval)
    - Learning rate: 2e-5, with linear warm-up
    - Optimiser: AdamW
    - Regularisation: Dropout (0.1), gradient clipping (1.0)

Multitask Models
- Encoder: Shared RoBERTa-base
- Heads:
    - Spam detection (binary)
    - Category classification (17-way)
    - Source prediction (~500-way)
- Loss weights:
    - Spam: $\alpha = 1.3$
    - Category: $\beta = 1.0$
    - Source: $\gamma = 0.5$
- Training setup: Same as single-task RoBERTa

4. Training Protocol

- Random seed: 42 (to ensure reproducibility)
- Evaluation metrics:
    - Spam detection: Accuracy, Precision, Recall, F1-score, ROC-AUC
    - Auxiliary tasks (Category, Source): Accuracy, Macro-F1
- Confidence weighting: Applied to reduce the influence of noisy weak labels
- Output storage: Pre-processed splits, metrics, and confusion matrices saved for reproducibility